

AI-generated

MARSNAV

Landing site selection and EVA navigation for the first humans on Mars

USER GUIDE & MISSION HANDBOOK

Version 2.1 · Windows desktop app · real NASA and USGS Mars data

NASA SPACE APPS CHALLENGE 2026

Challenge: Interplanetary Survival Guide: Martian Map

AT A GLANCE

Which way do we go, and can we always get home?

MarsNav is mission software for the first human crews on Mars. It helps a mission **choose where to land and build the base**, then **plan and fly every Marswalk (EVA)** from it, on real NASA and USGS data, with hard safety limits, a walk-home guarantee, and no Earth link needed.

66
candidate landing sites screened on real data: NASA's 48 Exploration Zones, 7 Starship sites, 11 landed spacecraft

5
named EVA routes in about 3 s on one laptop core, every one inside hard limits

0 / 40
simulated EVAs dipped into the reserve with MarsNav's closed loop, against 31 / 40 flying the filed plan

Offline
Mars mode: every map tile and dataset served from this computer; outside data requests refused and counted

117
automated tests pass (64 engine and closed-loop, 53 pipeline, science atlas and place); every build is tested on a clean Windows PC

0.086°
sun-angle error of the Mars clock against Perseverance's MEDA sensor, over 5,616 samples

21 / 30
real Perseverance detours where MarsNav's hazard map agrees the rover chose safer ground

1 click
to run: a native Windows app, no install, no Python, works on a fresh PC

Any point
of Mars, one click: ground, ice, minerals, air, where water boils, radiation, gravity, science and missions there

26
more whole-planet science layers: spectral, chemical, geophysical, atmospheric, radiation, with 74,250 mapped features

31
Mars missions catalogued, past, active, en route and planned, as of 28 October 2026

3
crew displays fed live: a helmet-visor HUD, a rover dashboard and a wrist cuff, with real maps

What you can do with it

Stage	Where	What it does
Choose the landing site	Landing mode	The whole of Mars in 3D from NASA / USGS data; every candidate scored on NASA's human landing-site criteria.
Know any place	Landing mode, EVA planner	Click any point on Mars or on a site: every data layer MarsNav holds about it, at once.
Choose the science	Landing mode, Science tab	26 more whole-planet layers, seven science objectives that rank every site, 31 missions, space weather.

Stage	Where	What it does
Commit the hub	Landing mode	Builds the chosen site into a full EVA site: elevation, hazards, imagery, weather, targets.
Plan the Marswalk	Plan	Hard limits first, then five named routes with oxygen, water and battery budgets, rehearsed in 3D.
Fly it	Execute	NASA SUITS telemetry, navigation without GPS, caution and warning, automatic re-planning.
Wear it	Crew displays	The same EVA on a helmet-visor HUD, a rover dashboard and a wrist cuff, with the ground under the crew.
Debrief	Review	Planned vs flown, event log, paper cue card, GeoJSON, validation on real rover data.
Go offline	Local / Mars mode	Every dataset on this computer at native resolution; then cut the Earth link.

START HERE

Find your way in

For judges: the two-minute path

- Download **MarsNav-Windows.zip** from `github.com/Aryan221224/MarsNav/releases/latest`, unzip it and double-click **MarsNav.exe** (Chapter 2).
- Click **Tour** in the top bar: a guided, seven-step walkthrough of the EVA planner (Chapter 12).
- Click **Landing mode** and click anywhere on Mars: every data layer at that point appears (Chapter 5). Then pick **Arcadia Planitia AP-1** and press **Assess landing site** (Chapter 4).
- In the **Science** tab, choose **Groundwater and the boxwork of Mount Sharp**, the challenge's own example, and press **Rank sites for it** (Chapter 6).
- In the EVA planner, press **Execute**, **Run**, then **HUD**: the helmet visor, rover dashboard and wrist cuff follow the EVA (Chapter 10).
- Open **Local mode** and switch on **Mars mode**: the app keeps working with the internet cut (Chapter 7).
- Read Chapter 16 for the validation against Perseverance and Chapter 17 for how MarsNav answers the challenge.

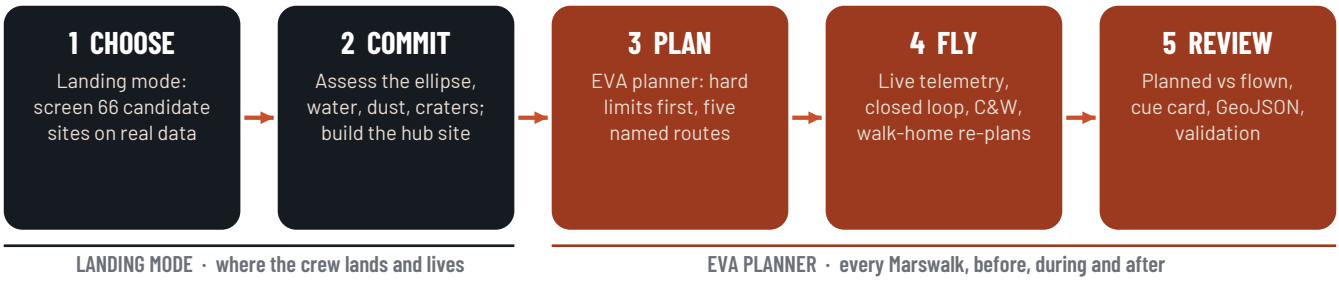
Who this guide is for

If you are...	Read	You will learn
A judge or reviewer	At a glance, Chapters 2, 12, 16, 17	What MarsNav is, how to see it working in minutes, and the evidence behind it.
A mission planner or crew member	Chapters 4-13	Choosing and committing a landing site, then planning, flying and debriefing an EVA on the crew's displays.
Showing it offline	Chapter 7	Downloading the data and running in Mars mode with no Earth link.

If you are...	Read	You will learn
A scientist or engineer	Chapters 5, 6, 14-16, 20	The data at any place, the science atlas, the algorithms, the validation and the honest limits.
A developer	Chapters 3, 18, 19, 21	How the app is put together, built, tested and fixed.

The mission journey

MarsNav follows the mission from orbit to the end of each Marswalk. Landing mode covers the first two stages and the EVA planner the last three; a hub committed in Landing mode becomes a full EVA site in the planner.



In this guide, **bold** marks buttons and menus as they appear in the app, `monospace` marks files and commands, and numbered red markers on a screenshot match the table under it.

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CHAPTER 01 · THE PROBLEM AND THE IDEA

Why MarsNav

The first humans on Mars will make two decisions that no map app can make for them: where to put the base, and which way to walk from it, every day, with a fixed supply of air and no one on Earth fast enough to help. MarsNav is built for both.

The operating problem

- **Light delay.** A radio signal takes 4 to 24 minutes each way. Mission control can advise, but the crew must decide on the spot.
- **No GPS.** Position comes from dead reckoning, the sun, landmarks and a beacon at the habitat, so every position has an error that grows with distance.
- **Finite consumables.** Suit oxygen, cooling water, CO_2 scrubber capacity and battery set hard ceilings. NASA's walkback rule says the crew must always be able to walk home, even if the rover fails.
- **Terrain that has already bitten.** Spirit was trapped in soft soil that looked firm; Opportunity was stuck for six weeks in a wind ripple. A boot meets slopes and steps that a 20 m elevation model cannot show.
- **Choosing the hub.** NASA's 2015 workshop proposed 48 Exploration Zones and SpaceX studied candidate Starship sites. Each must clear engineering limits and offer water ice, safe ground and science.

What MarsNav does differently

- **Hard limits before optimisation.** Unsafe ground, cliffs, keep-outs and anything beyond the walk-home budget are removed first. Only then are routes optimised, so no weight setting can trade safety away.
- **A walk-home guarantee inside the search.** Every partial route is dropped the moment the crew could not walk home with what is left. If no plan can work, MarsNav shows the point of no return and the resource that runs out there, rather than quietly relaxing a rule.
- **The ground a boot meets.** Walking-scale slope and step height from 1 m HiRISE stereo models, a ripple detector, and soft soil priced by its worst cases (CVaR), not its average.
- **A closed loop with the hardware.** During the EVA it reads suit telemetry in NASA's own SUITS format, learns how hard the crew is really working and how fast the suit is draining, and re-plans before the margin is gone.
- **Honest about what it does not know.** Data confidence on every route, assumptions tagged `[ASSUME]` in the code, and validation against Perseverance's own records, including the result that did not go our way.
- **Everything about a place, one click away.** Any point on Mars, or on a site, shows every layer MarsNav holds there: ground, ice, minerals, air, where water boils, radiation, gravity, science, missions and the best data.
- **Science first.** Twenty-six more whole-planet layers grouped by the kind of measurement, seven science objectives that rank every candidate site, the missions past, present and future, and space weather.

- **On the crew's own displays.** The same EVA, live, on a helmet-visor HUD, a rover dashboard and a wrist cuff, with the orbital map under the route and the facts about the ground the crew stands on.
- **Built for Mars, so it runs without the internet.** Local mode puts every dataset on the computer; Mars mode proves it by refusing every outside request.



Figure 1. *The EVA planner on real Jezero crater data: the EVA set-up (left), the composite safety-hazard map with the planned route to the Amalik delta front, where Perseverance cored Mageik (centre), the five named routes (right) and the elevation, walk-home margin and sun profile along the route (bottom).*

CHAPTER 02 · TWO MINUTES FROM DOWNLOAD TO MARS

Quick start

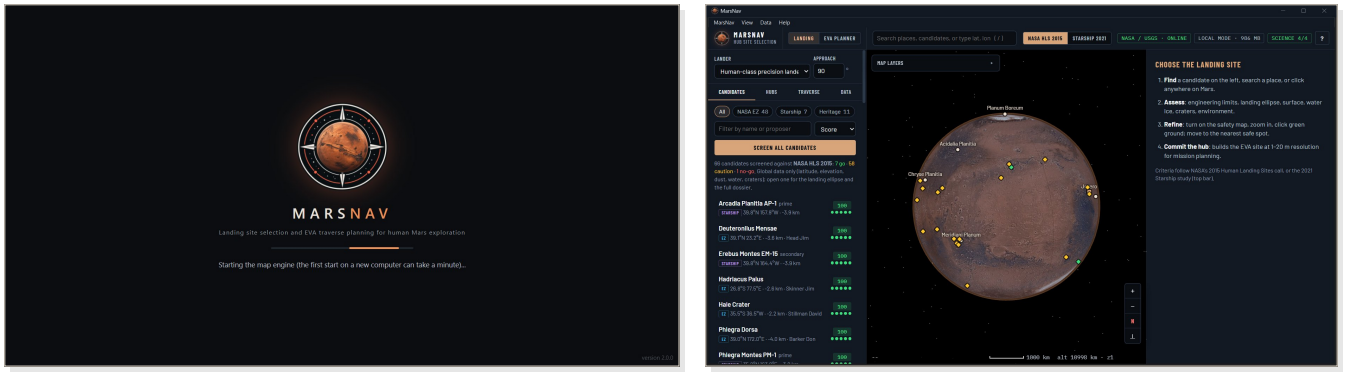
MarsNav is a portable Windows program: one folder, one double-click. Nothing is installed and nothing needs admin rights, so it runs the same on a judge's laptop as on the development machine.

What you need

Item	Requirement
Computer	Windows 10 or 11, 64-bit. 4 GB RAM (8 GB recommended).
Disk	About 300 MB for the program (a 155 MB download), plus room for data the app downloads (the tile cache is capped at 20 GB; Local mode packs are optional and sized before you download).
Window engine	Microsoft Edge WebView2, which is part of Windows 10 and 11. On a PC without it, MarsNav opens in a Microsoft Edge app window instead (no tabs, no address bar).
Internet	Optional. The Jezero EVA planner works fully offline from the data shipped in the folder. Landing mode streams whole-planet data the first time you look at a place, or uses Local mode packs.
Python, Node, admin rights	Not needed.

Run it

1. Download **MarsNav-Windows.zip** from the latest release, `github.com/Aryan221224/MarsNav/releases/latest`. Every release is built and tested on a clean Windows machine before it is published (Chapter 18).
2. Unzip it anywhere you can save files: your Desktop, Documents, another drive such as D:\, or a USB stick. Not inside C:\Program Files.
3. Open the **MarsNav** folder and double-click **MarsNav.exe**.
4. The first time on a new PC, Windows may show **"Windows protected your PC"** because the program is not code-signed. Click **More info**, then **Run anyway**. This happens once.
5. The start-up screen appears while the map engine starts (a few seconds; up to a minute the very first time while Windows checks the files). The EVA planner then opens on Jezero crater.
6. Optional: **MarsNav** menu, **Add desktop and Start menu shortcuts**.



Start-up screen Landing mode, a few seconds later

Figure 2. *MarsNav starting on a fresh PC: the logo start-up screen while the local map engine starts, then the app in its own window, with its own icon in the title bar and taskbar.*

What is in the folder

Item	What it is
`MarsNav.exe`	The program. Double-click to run.
`MarsNav_runtime\`	Its libraries (Python, numpy, scipy, GDAL / rasterio, WebView2 bindings). Keep it next to the exe.
`app\`	The interface and the planning engine, plus the Jezero site data (works offline).
`planet\, pipeline\`	The whole-planet engine and the data pipeline used to build new sites.
`MarsNav_User_Guide.pdf`	This guide (also Help, User guide).
`START HERE.txt`	A one-page version of this chapter.
`data_cache\`	Created on first run: everything MarsNav downloads or builds (map tiles, science layers, Local mode packs, built hub sites, reports, logs).

> **Keep the folder together**
> MarsNav.exe needs the folders next to it. Move or copy the whole MarsNav folder, never the .exe alone. To remove MarsNav, delete the folder (and the shortcuts if you added them).

Your first ten minutes

1. Press **Tour** in the top bar for a guided walkthrough of the EVA planner.
2. Press **Plan EVA** and compare the five route cards on the right.
3. Switch to **Execute**, set the speed to 120x and press **Run**; then press **Suit leak** and watch the warning and the automatic walk-home.
4. While the EVA runs, press **HUD** on the map: the helmet visor opens, with the rover dashboard and the wrist cuff one key away (1, 2, 3).
5. Click any point of the map: a card shows the site's terrain there and every planet-wide layer.
6. Press **Landing mode**, click anywhere on Mars, then choose a candidate and press **Assess landing site**.

7. Open **Local mode** and switch on **Mars mode**.

CHAPTER 03 · WINDOW, MENUS AND FILES

The MarsNav desktop app

MarsNav runs in its own native window, not in a web browser. Behind the window, a supervised local engine streams the planet, runs the heavy computations and keeps itself healthy.



Figure 3. *The MarsNav window (EVA planner). Numbers mark the parts described below.*

#	Part	What it shows or does
1	Menu bar	MarsNav, View, Data and Help menus (table below), in the same console colours as the window.
2	MarsNav and site	The site being planned (here Jezero crater).
3	Module switch	Landing (where the crew lands and lives) and EVA planner (plan, fly and review a Marswalk).
4	EVA phase	Plan, Execute and Review.
5	Mars clock	UTC, mission sol, local mean solar time and sun elevation; wider windows also show the season (Ls) and the one-way radio delay to Earth.
6	Annunciators	The EVA phase and telemetry source (PRE-EVA, EVA · SIM TLM, EVA · LIVE TLM, POST-EVA), then the W / C / A counters: warnings, cautions, advisories.
7	Tour	The guided two-minute walkthrough (Chapter 12).
8	Left panel	EVA set-up in Plan, telemetry and controls in Execute, exports and the event log in Review.

#	Part	What it shows or does
9	Map tools	Zoom, fit the route, 3D view, Drive (the first-person drive window) and HUD (the crew displays, Chapter 10).
10	Layers	Shows the layer on the map; click it (or press L) for the eleven layers and four overlays.
11	Right panel	Route cards in Plan, alerts and the closed loop in Execute, validation in Review.
12	The route	The selected route in white, alternatives dashed, no-go areas hatched.
13	Legend	What the active layer means and the data it comes from (DERIVED, MODELLED or [ASSUME] tags).
14	Timeline	Elevation, walk-home margin and sun angle along the route; a cursor follows the crew in Execute.

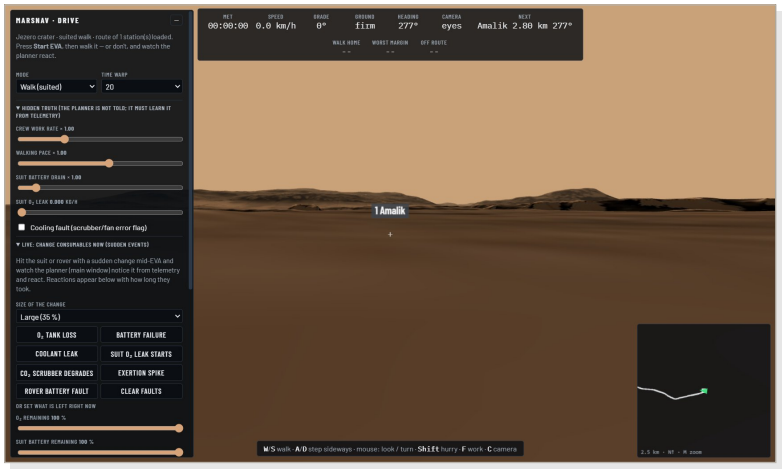
The menus

Menu	Command	What it does
MarsNav	Landing mode (hub site selection)	Open Landing mode.
	EVA planner (Jezero traverse planning)	Open the EVA planner.
	Jezero east demo site	The planner on a site built from 20 m data only (Chapter 13).
	Add desktop and Start menu shortcuts	What an installer would add, for this copy of MarsNav.
	Quit	Close MarsNav and stop everything it started.
View	Reload page (F5) · Back · Full screen (F11)	Window controls.
Data	Local mode: download the data for offline use…	Opens the Local mode panel (Chapter 7).
	Rebuild Jezero data from raw NASA / USGS sources…	Re-runs the data pipeline in its own console window (Chapter 18).
	Full rebuild (also the Jezero 1 m DTMs)…	The same, with three optional HiRISE elevation models.
	Open the data folder · Open saved reports	Explorer on data_cache and on saved dossiers.
Help	User guide (PDF)	Opens this guide.
	Engine status	Map engine health, memory, raster workers and free disk.
	Open the log folder · About MarsNav	Logs for troubleshooting; version and window engine.

Windows, dialogs and files

- **Extra windows.** The 3D drive window, the crew displays and printable dossiers open as further MarsNav windows. The drive window and the crew displays talk to the main window live, so you can walk the route in one, watch the visor in another and the planner react in the third.
- **Saving.** Every export (cue card, GeoJSON, dossier JSON) opens the normal Windows **Save As** dialog. Dossiers are also kept in `data_cache\reports`.
- **Links** to NASA abstracts and data sources open in your web browser.

- **One copy at a time.** Starting MarsNav again brings the open window to the front.
- **It remembers** the window size, position and the page you were on, and whether the layer list is open.
- **Explanations on request.** Panels show values, not paragraphs: an **i** next to a section heading opens a short explanation of that section.
- **Closing** the window stops the map engine and every helper process. If a Local mode download is running, MarsNav asks first; downloads resume where they stopped.

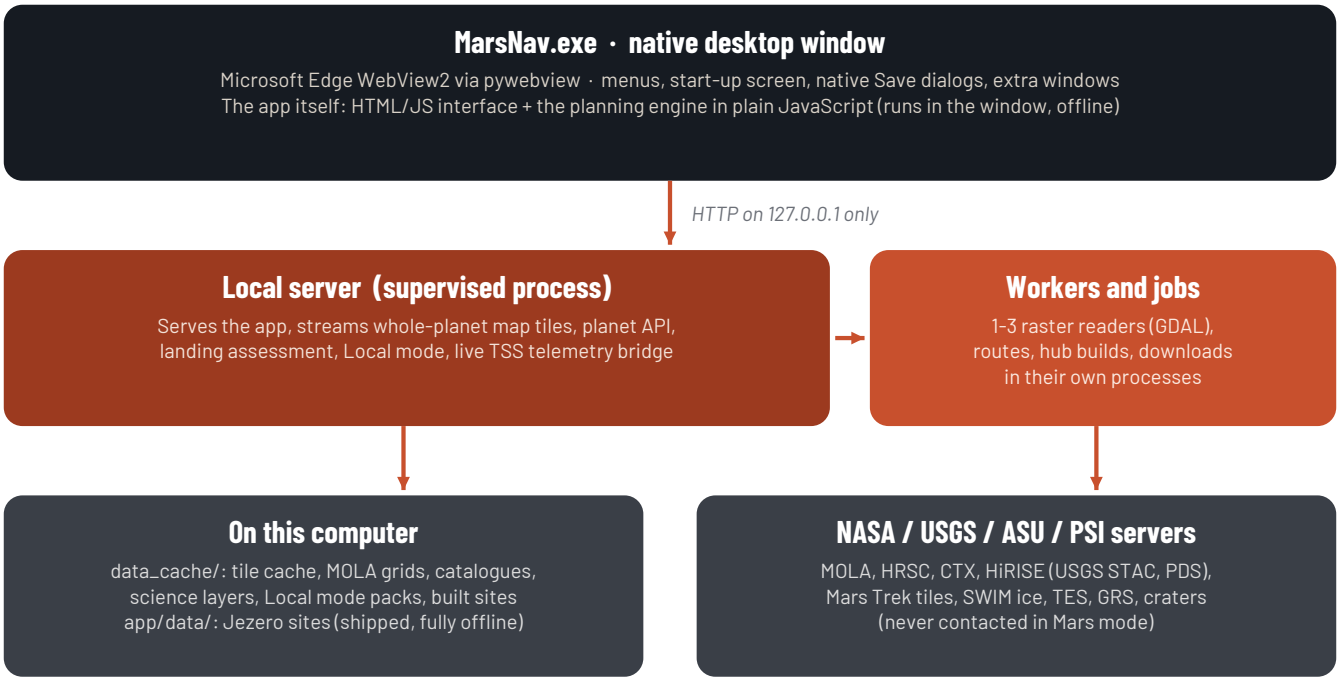


3D drive window | Printable hub dossier

Figure 4. *Further windows: the first-person drive window (left), whose movement becomes suit telemetry for the main window, and a hub-site dossier ready to print or save as PDF (right).*

Behind the window

The window is the visible part of a small, robust system. A supervisor starts the local map engine in its own process, checks it every five seconds on a private health port, and replaces it within seconds if it ever stops answering. Raster reads from NASA and USGS servers run in one to three worker processes, and long jobs (routes, hub builds, downloads) in their own processes, so a slow server can never freeze the app. Everything runs inside a Windows job object: closing MarsNav ends every process it started. The built-in self-test kills a worker and the server on purpose under load and checks that everything recovers (Chapter 16).


MarsNav.exe · native desktop window
Microsoft Edge WebView2 via pywebview · menus, start-up screen, native Save dialogs, extra windows
The app itself: HTML/JS interface + the planning engine in plain JavaScript (runs in the window, offline)
HTTP on 127.0.0.1 only
Local server (supervised process)
Serves the app, streams whole-planet map tiles, planet API, landing assessment, Local mode, live TSS telemetry bridge
Workers and jobs
1-3 raster readers (GDAL), routes, hub builds, downloads in their own processes
On this computer
data_cache/: tile cache, MOLA grids, catalogues, science layers, Local mode packs, built sites
app/data/: Jezero sites (shipped, fully offline)
NASA / USGS / ASU / PSI servers
MOLA, HRSC, CTX, HiRISE (USGS STAC, PDS), Mars Trek tiles, SWIM ice, TES, GRS, craters (never contacted in Mars mode)

CHAPTER 04 · CHOOSE WHERE THE CREW LANDS AND LIVES

Landing mode

Landing mode is where the mission picks the hub: the place the crew lands, sets up the habitat and starts every EVA from. It shows the whole of Mars in 3D on real NASA and USGS data, screens every candidate site against NASA's human landing-site criteria, and builds the chosen hub into a full EVA site.

Open it with **Landing** in the top bar (the module switch next to the MarsNav name), or the MarsNav menu. The globe is streamed the way a flight simulator streams the Earth: only the map tiles in view are loaded, each at the best resolution NASA and USGS have mapped there, from 3.7 km for the whole planet down to 1 m HiRISE elevation and 25 cm imagery close up.

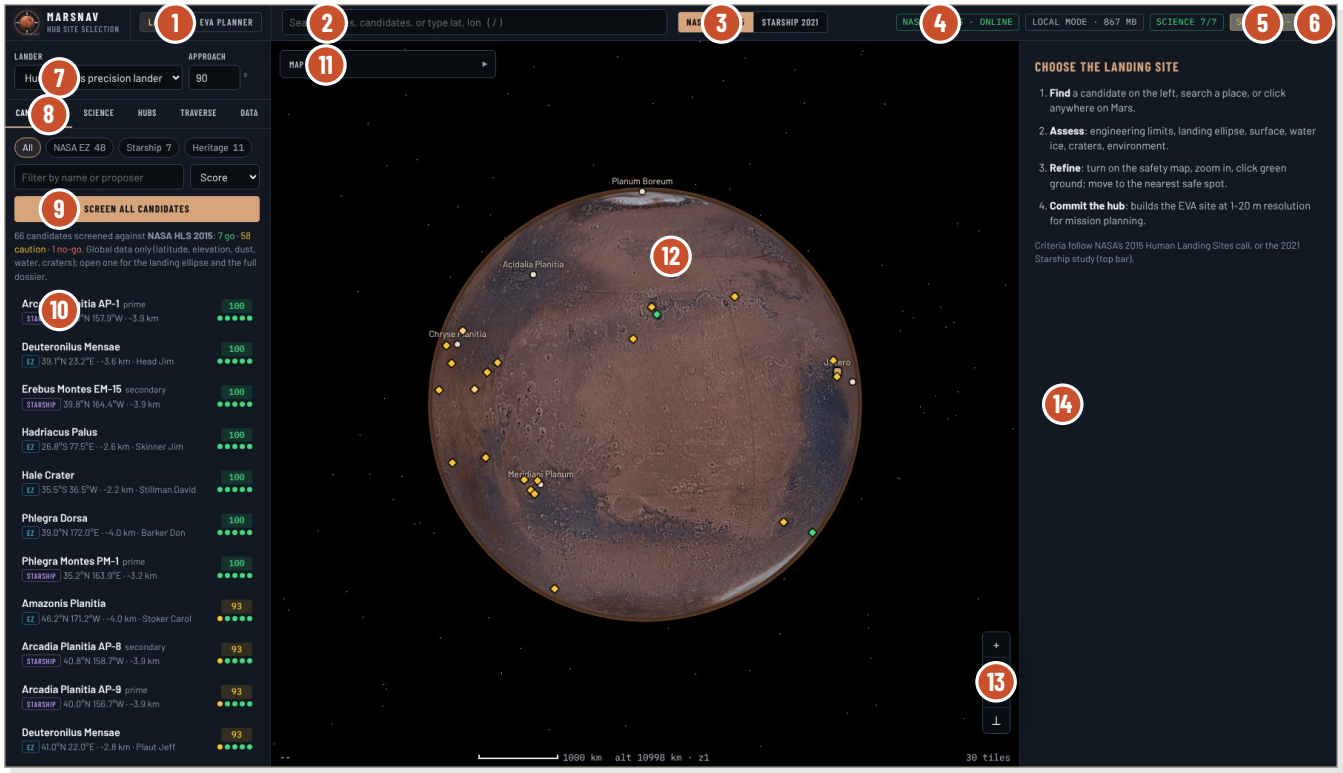


Figure 5. *Landing mode on first opening: 66 candidate sites, screened on real data, on the globe and in the list.*

#	Part	What it does
1	Module switch	Landing (this page) and EVA planner.
2	Search	Place names, candidate sites, or coordinates such as 18.4N 77.5E (press / to jump here).
3	Criteria set	NASA HLS 2015 (Human Landing Sites call) or Starship 2021 (Golombek et al.).
4	Annunciators	Connection to NASA / USGS, Local mode data on this computer (click to open Local mode), and SCIENCE n/7: how many installable science layers are on this computer.

#	Part	What it does
5	Space weather	SPACE WX: NASA DONKI space weather at Mars, quiet, watch or storm. Click it for the digest (Chapter 6).
6	Help	The keyboard shortcuts and how to read the scores.
7	Mission	Lander class and approach azimuth. They set the size and orientation of the landing ellipse.
8	Tabs	Candidates, Science (objectives and missions, Chapter 6), Hubs (sites already built), Traverse (long drives) and Data (science layers, offline download, sources).
9	Screen all candidates	Re-scores every candidate on global data in a second or two.
10	Candidate list	Each site with a 0-100 score and five dots: latitude, elevation, dust, water, craters.
11	Map layers	Base imagery, overlays and the whole-planet science atlas (press L).
12	The globe	Drag to move, wheel to zoom, right-drag to tilt, double-click to fly. Click any point for every data layer there (Chapter 5). Dots are coloured by verdict; labels that would overlap give way.
13	View controls	Zoom, north up, tilt.
14	Site dossier	The selected place: score, verdict, criteria, actions, and all the data at that point.

Step 1 · Find a candidate

- **NASA EZ**: the 48 Exploration Zones proposed at NASA's First Landing Site / Exploration Zone Workshop for Human Missions to the Surface of Mars (2015), each with its proposer and links to the abstract and presentation.
- **Starship**: 7 candidate sites from the SpaceX Starship study (Golombek et al. 2021).
- **Heritage**: 11 places where spacecraft have landed, for comparison.
- Every candidate is screened when the page opens. Filter by name or proposer, sort by score, name or latitude, or simply click anywhere on Mars to screen that point and see every data layer there (Chapter 5).

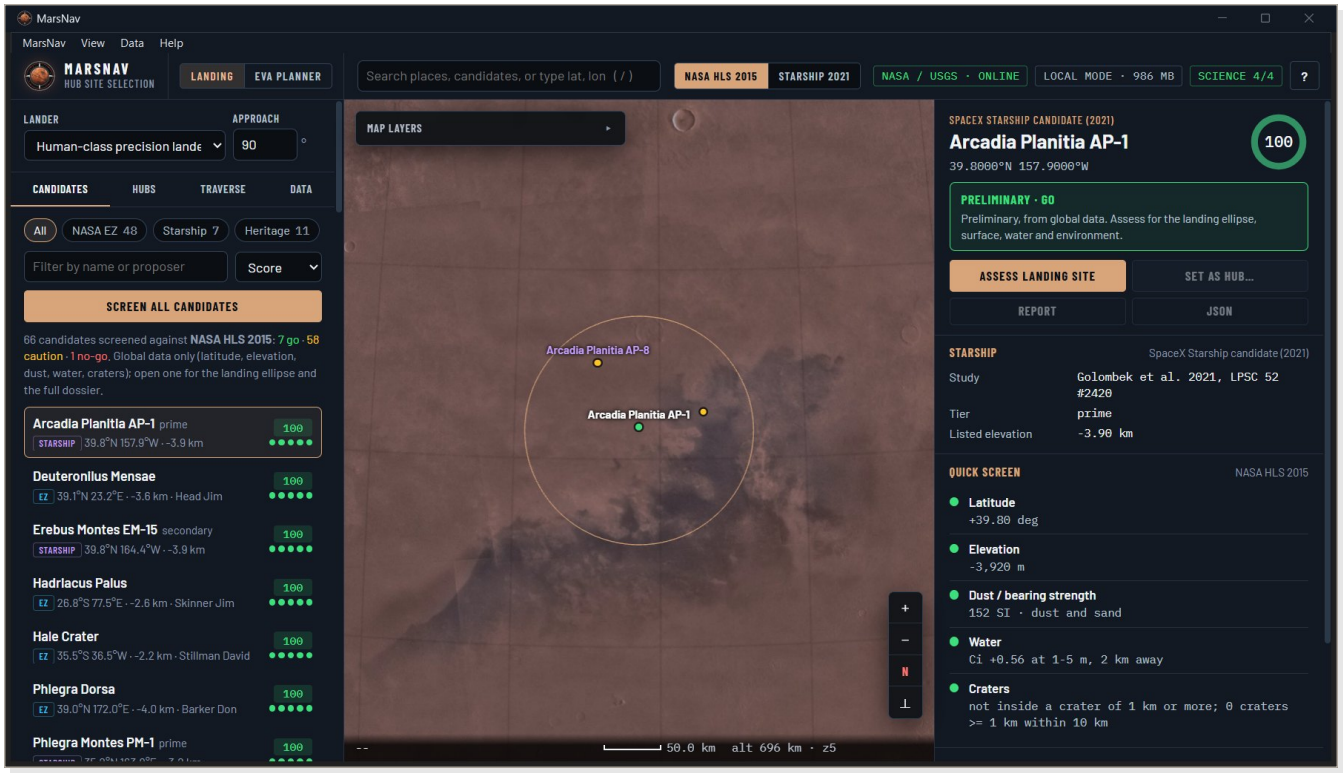


Figure 6. *Arcadia Planitia AP-1 (SpaceX's prime Starship site) selected: the 100 km zone is drawn around it and the dossier shows a preliminary screen from global data, before the full assessment.*

Step 2 · Assess the landing site

Assess landing site (or the A key) runs the full assessment for the chosen lander and approach azimuth. The engine builds the best elevation mosaic over the landing ellipse (HiRISE 1 m where it exists, then CTX 20 m, HRSC 200 m and MOLA 463 m), classifies the touchdown point and every part of the ellipse, and evaluates each criterion on the real data. The result is a score out of 100 and a verdict:

Verdict	Meaning
GO	Every criterion is met.
CAUTION	Usable with mitigations: at least one criterion is amber, or a non-critical one is red.
NO-GO	Latitude, elevation or the landing ellipse fails. The crew should not land there.

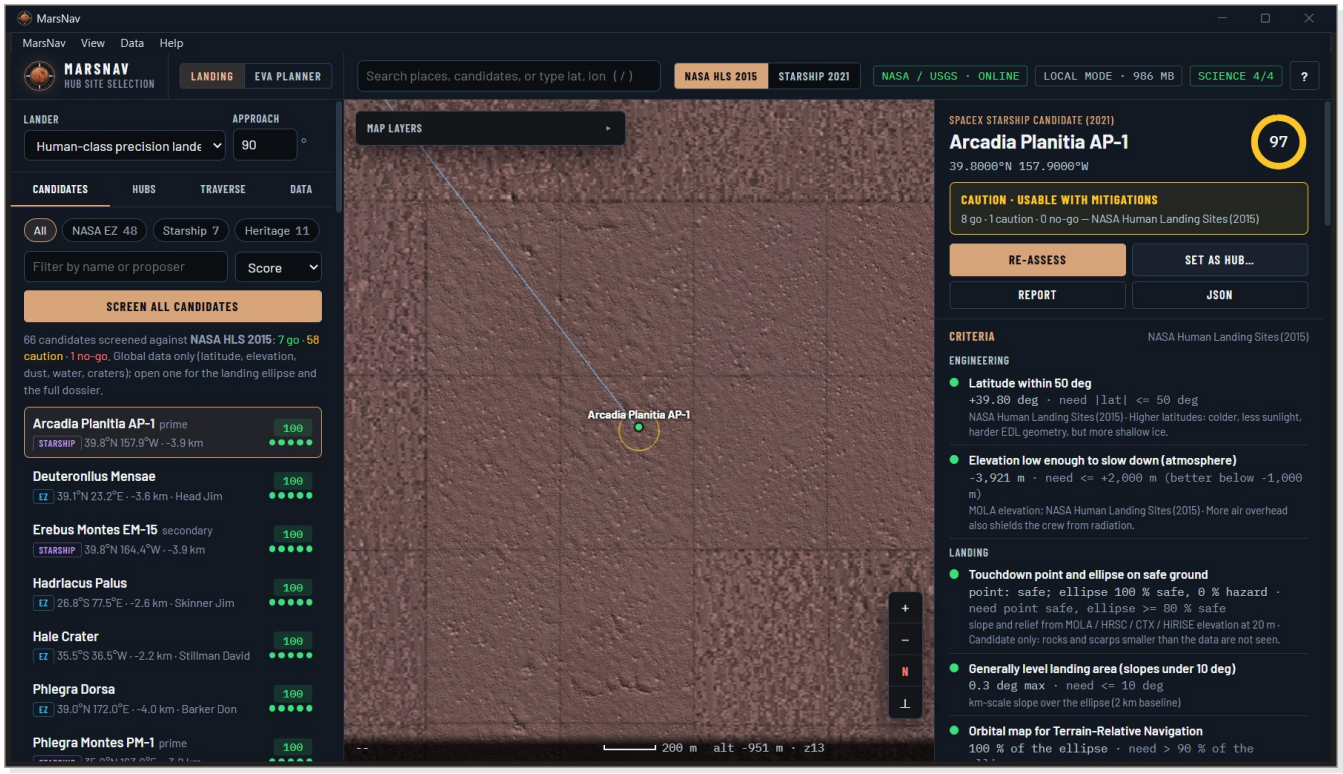


Figure 7. *AP-1 assessed for a human-class precision lander: score 97, CAUTION (8 criteria green, 1 amber: no 1 m HiRISE elevation yet over the ellipse). The 150 m ellipse is drawn on the CTX 5 m mosaic.*

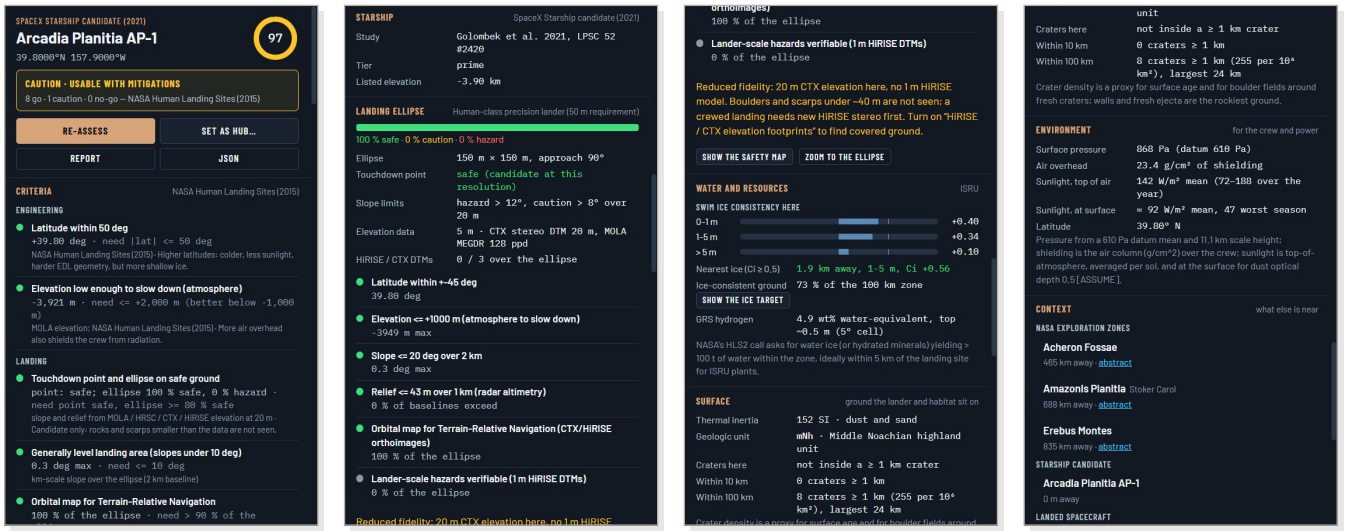
The criteria

Criterion	NASA HLS 2015	Starship 2021	Data used
Latitude	≤ 50° (amber above 40°)	≤ 40° (amber above 35°)	position
Elevation	≤ +2 km (best below −1 km)	≤ −2 km (best below −3 km)	MOLA MEGDR
Landing ellipse	point safe, ≥ 80 % of the ellipse safe	same	slope and relief on the best elevation
Level ground	slopes under 10° over 2 km	under 5°	elevation mosaic
TRN map	orbital images over > 90 % of the ellipse	same	CTX / HiRISE footprints
1 m verification	HiRISE elevation over > 50 %	same	USGS STAC catalogue
Dust	thermal inertia ≥ 100 (no thick fine dust)	same	TES (Putzig & Mellon 2007)
Water	ice (Ci ≥ 0.5, 0-5 m) within 5 km; amber within 100 km	same	SWIM; GRS if SWIM is not installed
Craters	not in a crater or on a crater wall; few craters within 10 km	same	Robbins & Hynek database

The dossier

Below the verdict, the dossier explains every number, section by section, with its data source:

- **Criteria**, grouped (engineering, landing, resources, surface), each with the measured value, the limit and a note.
- **Landing ellipse**: its size and orientation, the share that is safe / caution / hazard / no-go, the class of the touchdown point and the elevation data under it. If the point is not safe, **Move to nearest safe spot** fixes it.
- **Water and resources**: SWIM ice consistency at 0-1 m, 1-5 m and deeper, the nearest ice-consistent ground (**Show the ice target** draws a line to it), and the GRS hydrogen value.
- **Surface**: thermal inertia (dust, sand or rock), the geologic unit, and craters nearby.
- **Environment**: surface pressure, the air overhead (radiation shielding) and sunlight through the year.
- **Context**: the nearest Exploration Zones, Starship sites, landed spacecraft and named features.
- **Science along the way**: the catalogued minerals, ice landforms, active slopes and mission sites within 100 km, the nearest of each, and a planetary-protection warning when a candidate Special Region is near (Chapter 6).
- **Science objectives here**: the seven objectives scored on the evidence within 100 km.
- **Chemistry, gravity and radiation**: the GRS elements, albedo, the gravity anomaly, the cosmic-ray dose for the air overhead and space weather now; then the **data sources** with citations.
- Below the dossier, **All data at this place** stays open for the same point, folded (Chapter 5).



Verdict and criteria | Landing ellipse | Water and surface | Environment, context

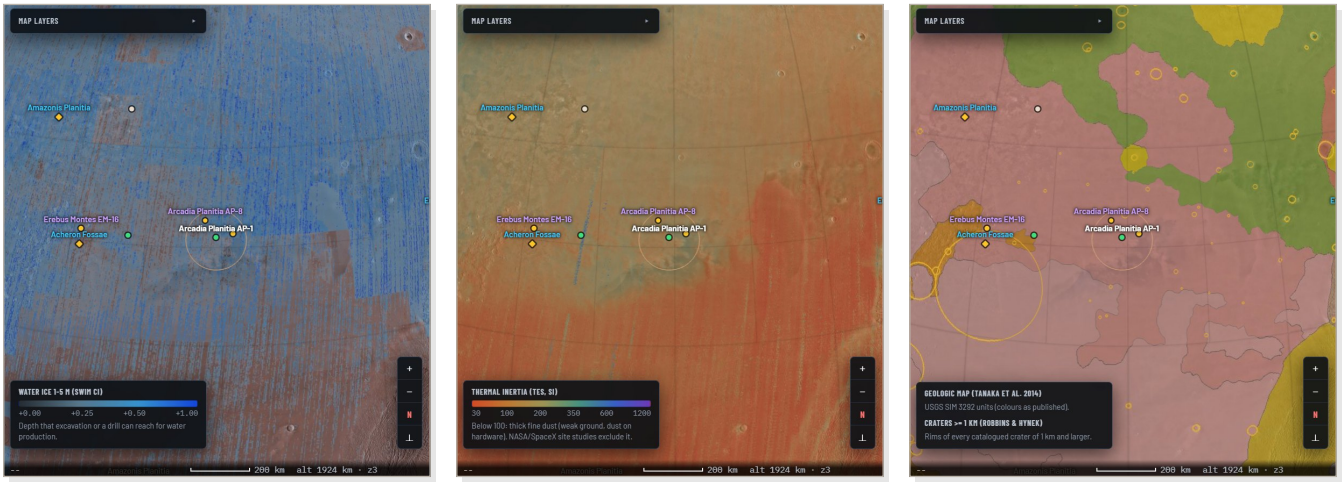
Figure 8. *The AP-1 dossier: shallow water ice 1.9 km from the touchdown point (SWIM Ci +0.56 at 1-5 m), 73 % of the 100 km zone ice-consistent, 100 % of the ellipse safe at the available resolution, and a clear warning about what the data cannot show.*

Step 3 · Refine with the map layers

Map layers (top left of the globe, or L) sets the base imagery and the overlays drawn on top. The legend in the bottom-left corner always says what is shown and where it comes from.

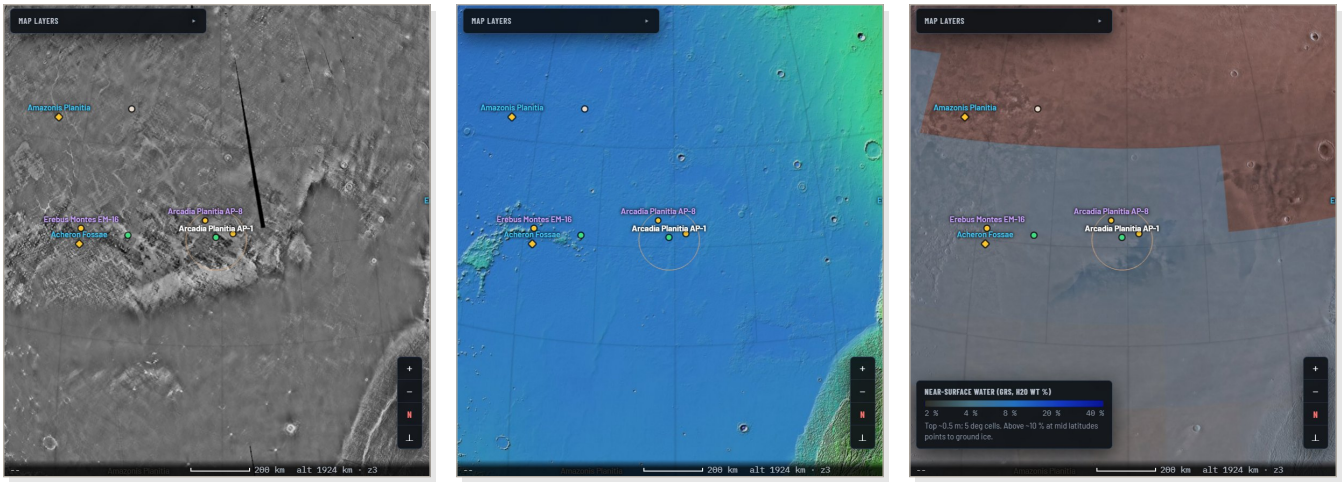
Layer	What it tells the mission
Best available imagery	Colour from orbit, then the CTX 5 m global mosaic, then HiRISE down to 25 cm, as you zoom in.

Layer	What it tells the mission
THEMIS night / day infrared	Night IR: rocky ground stays warm and bright. Day IR at 100 m: morphology everywhere.
MOLA colour elevation, Viking colour	Topography at a glance; the classic colour mosaic.
Landing safety (for the chosen lander)	Green safe (strong green verified at 1-2 m), amber caution, red hazard, purple no-go. It sharpens as you zoom, because the elevation under it does.
Water ice (SWIM)	Blue where every orbital dataset agrees with shallow ice, at 0-1 m, 1-5 m or deeper.
Thermal inertia (TES)	Orange = fine loose dust (weak ground, dust on hardware), blue = rock.
Near-surface water (GRS)	Water-equivalent hydrogen in the top half metre (5° cells).
Geologic map, craters ≥ 1 km	USGS SIM 3292 units in their published colours; rims of 384,343 catalogued craters.
Science atlas · whole planet	Twenty-six more layers in nine groups: minerals, elements, water and ice landforms, thermal, topographic, gravity, air pressure, dose, sunlight, present-day activity and mission sites (Chapter 6).



Water ice (SWIM, 1-5 m) Thermal inertia (TES) Geologic map + craters

Figure 9. *Science overlays around Arcadia Planitia, about 2,000 km across: shallow ice is consistent across the plains (left), the surface is sand and duricrust rather than deep dust (centre), and the geologic units and crater rims (right).*



THEMIS night infrared | MOLA colour elevation | Near-surface water (GRS)

Figure 10. *Base imagery and the GRS hydrogen map over the same region.*

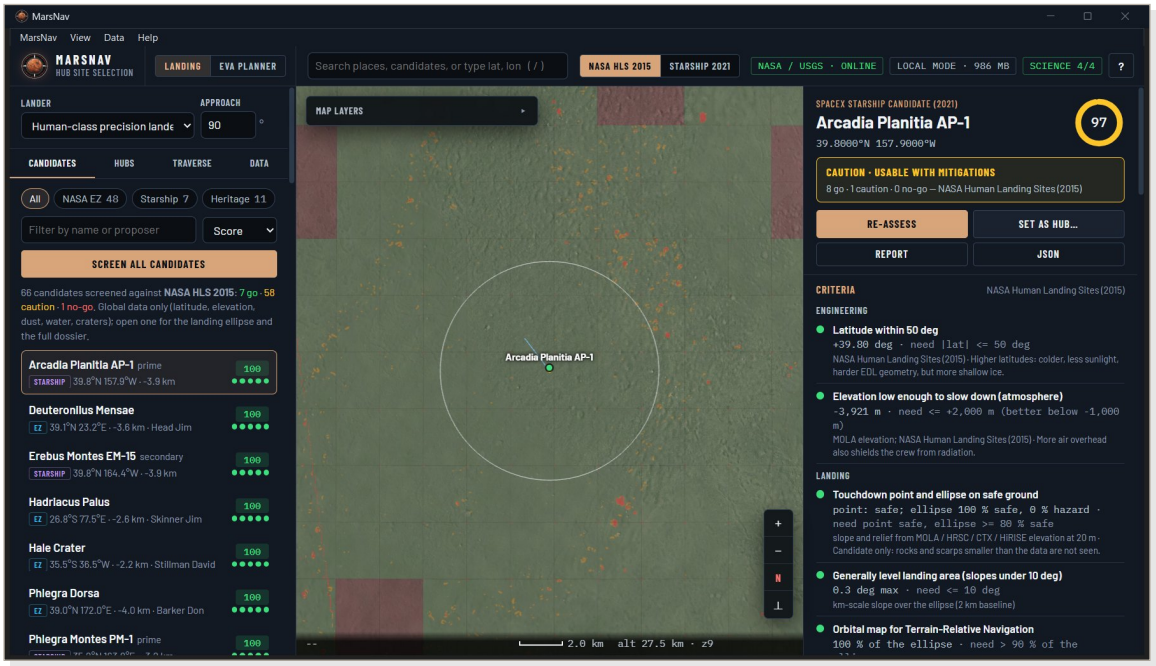
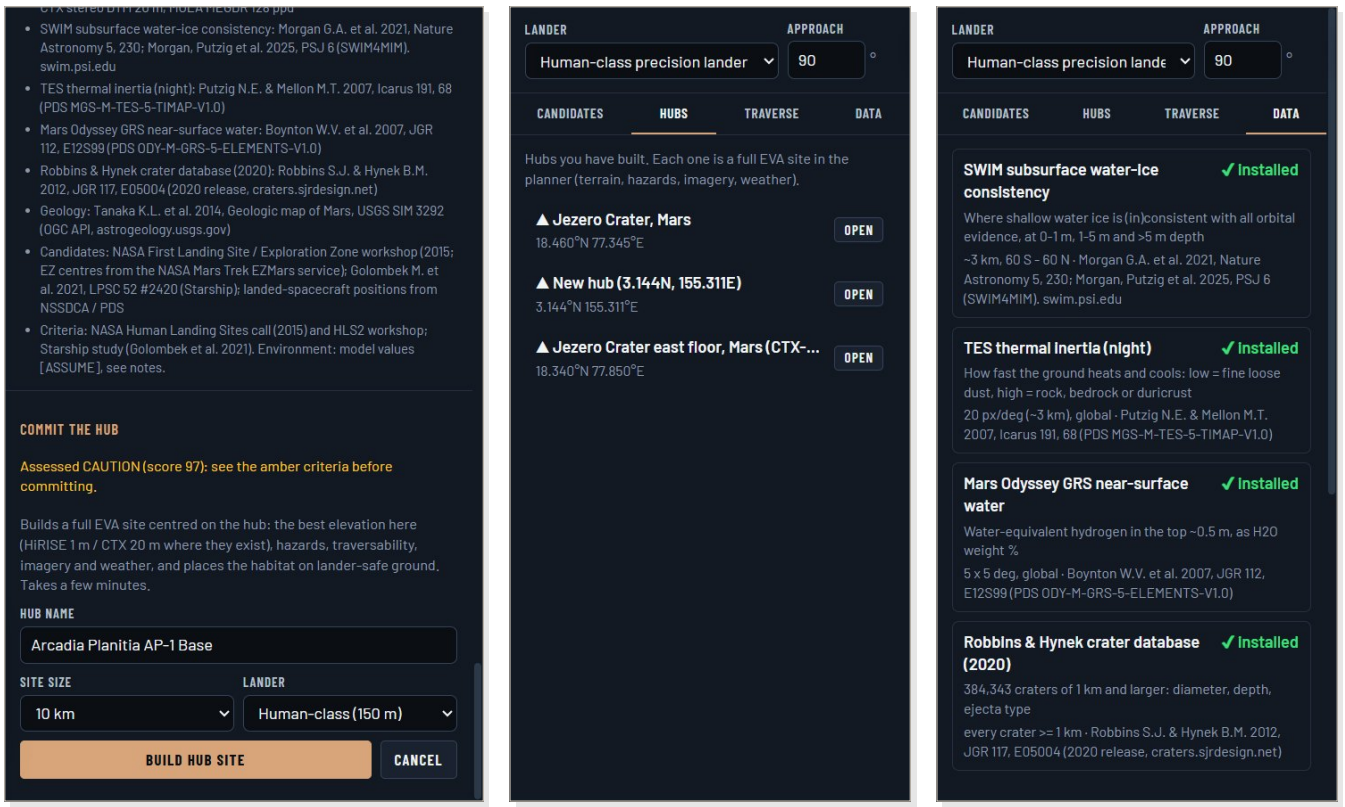


Figure 11. *The landing-safety overlay for a human-class lander, zoomed to about 30 km: green ground is safe at the resolution shown. Where 1 m HiRISE elevation exists the map reaches lander scale and the green becomes 'verified'.*

Step 4 · Commit the hub

Set as hub... opens the commit card. It repeats the verdict and asks for the hub's name, the size of the EVA site (6 to 24 km) and the lander. **Build hub site** then builds a complete EVA site centred on the hub, with the habitat exactly there. It takes a few minutes and the log shows each step. When it is done, **Open ... in the EVA planner** takes you to the new site, which also appears in the **Hubs** tab and on the globe.



Commit the hub | Hubs tab | Data tab

Figure 12. *Committing AP-1 as a hub (left); the hubs already built, each a full EVA site (centre); the science layers installed once on this computer, each with its source (right).*

What a new hub site gets

- The same pipeline as Jezero: a 10 m operations grid with slope, steps, roughness, sand, soft soil, cliffs, landmarks and science layers, walk-home fields, 3D, the drive window and the closed loop.
- Elevation: CTX 20 m where it exists, blended into MOLA; HiRISE 1 m models over the operations area (up to eight, those covering the hub first); otherwise 463 m MOLA with walking-scale slopes calibrated from Jezero's 1 m data.
- Imagery: a CTX 5 m mosaic, and 2 m HiRISE detail where 1 m elevation exists.
- Weather: Jezero's MEDA climatology moved to the new latitude and elevation (sunlight through the Martian year, the CO_2 frost point, hydrostatic pressure).
- Named features inside the area become science targets; otherwise survey stations are placed around the hub.
- To remove a hub you built, open it in the EVA planner and press **Delete site** (the Jezero sites are kept).

Report and JSON

Report opens a printable dossier, with a picture of the site, in its own window: press **Print / save as PDF**. **JSON** saves the complete assessment for other tools.

Traverse: long drives between sites

In the **Traverse** tab, add the selected point (and more), choose the traveller (crewed pressurised rover, robotic rover or suited walk) and press **Plan traverse**. The router finds the least-cost path on the real elevation, coarse to fine: slopes over the traveller's limit are impassable, and climbing costs more than descending. The result shows distance, climb, steepest slope, driving time in sols and an elevation profile with over-limit stretches in red.

Science layers and keyboard

Seven datasets are downloaded once (about 300 MB in all) and converted on this computer: SWIM ice consistency (Morgan et al. 2021, 2025), TES thermal inertia (Putzig & Mellon 2007), Mars Odyssey GRS hydrogen and elements (Boynton et al. 2007), TES albedo, the MRO GMM-3 gravity field and the Robbins & Hynek crater database (384,343 craters ≥ 1 km). Until they are installed, the banner over the globe offers to install them and the criteria that need them are judged from coarser data, and say so. Each layer installs on its own: if one server does not answer, the others still work. The feature catalogues of the science atlas are part of the app and need no download.

Keys	Action	Keys	Action
Drag	move the globe	Wheel, + / –	zoom (towards the cursor)
Right-drag, Shift+drag	tilt and turn	Double-click	fly there
Arrows · Q / E	pan · turn	PgUp / PgDn	tilt
N	north up	/	search
A	assess the selection	L	map layers
?	help	Esc	close a panel

CHAPTER 05 · EVERY DATA LAYER AT ONE POINT, ANYWHERE ON MARS

Everything at any place

Click any point on the globe, or any point of a built EVA site, and everything MarsNav holds about it appears at once: where it is, the time and sun there, whether a crew could survive and work there, the ground, water, minerals, science, the missions nearby and the best data covering it. No assessment has to run first.

In Landing mode

Click anywhere on the globe, or pick a candidate. The dossier first shows the quick screen against the criteria set and the best elevation at the current zoom, then **All data at this place**: the time there, live, and nine folding sections, the first four open. They stay below the dossier when you press **Assess landing site**. **JSON** saves them with the assessment (or alone, before one), and **Report** prints them; **HUD** opens the crew displays as a survey of the point (Chapter 10).

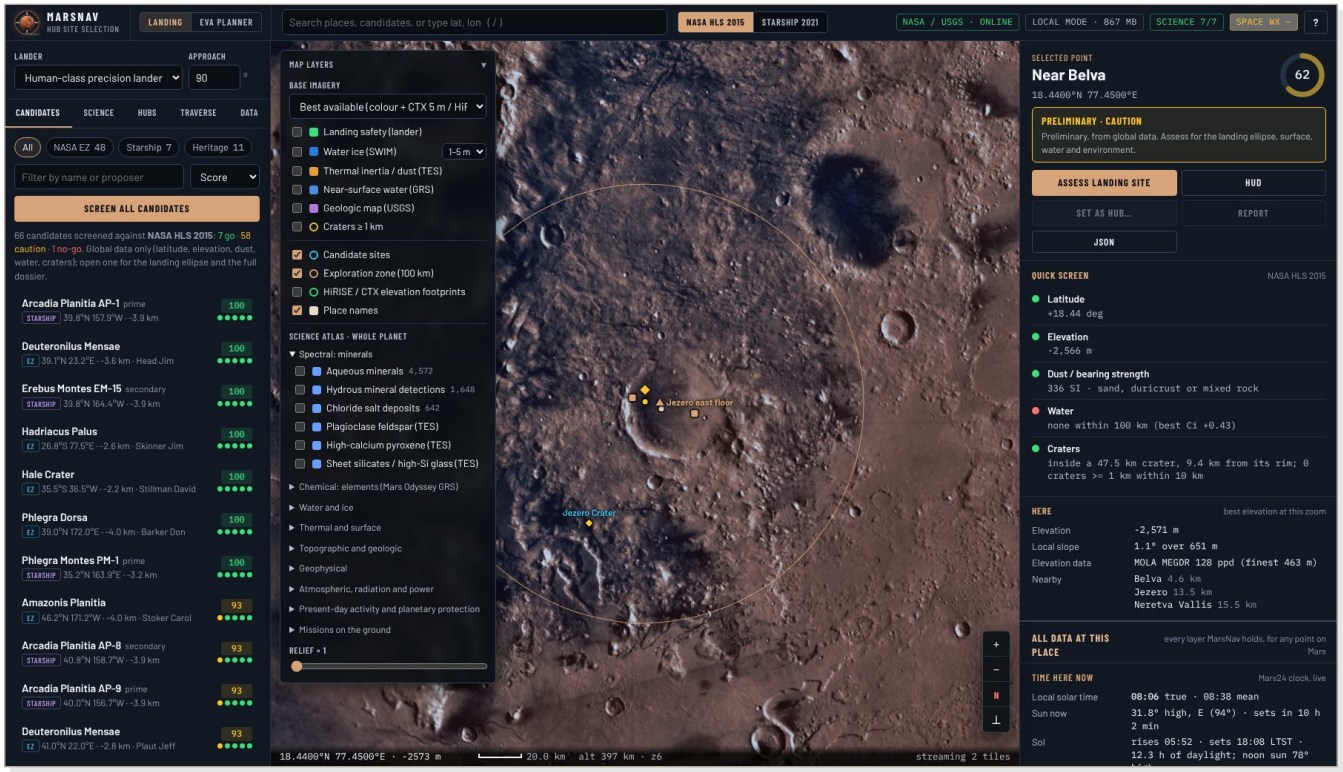


Figure 13. *A point clicked on the floor of Jezero crater: the quick screen against NASA's 2015 criteria, the elevation and slope at this zoom, and the start of All data at this place, with the local time and the sun.*



Time and where | Survival and ground | Water and composition | Science, missions, data

Figure 14. *The same point, section by section: local solar time, sun and season; the USGS chart and names; air pressure, the liquid-water window, dose and gravity; thermal inertia, geology, dust and roughness; SWIM ice at three depths and GRS hydrogen; minerals and elements; the catalogued features and objective scores; the missions and the elevation models covering the point.*

What each section holds

Section	What it holds	From
Time here now	Local true and mean solar time, the sun's height and direction, sunrise, sunset and daylight length, the noon sun, the season in each hemisphere, Ls and Mars Year, the dust-storm season, sunlight above the air, the Earth's distance and radio delay. Polar day and night are named.	Mars24 clock, JPL planet elements
Where	Position (east and west longitude), the USGS 1:5M chart quadrangle (MC-01 to MC-30), lowlands or highlands, MOLA elevation, named features, and the built EVA site the point lies in or the nearest hub.	IAU Gazetteer, USGS, MOLA
Survival here	Mean air pressure, whether pure liquid water can exist and at what temperature it boils, the air overhead as radiation shielding, the cosmic-ray dose per day and over a 500-sol stay, surface gravity, sunlight above the air and at the ground, space weather now.	MOLA, Curiosity RAD, MRO gravity, NASA DONKI
Ground	Thermal inertia and what it means (dust, sand, rock, or ice near the poles), the geologic unit, albedo, the crater the point is in, crater counts within 10 and 100 km, dust cover and roughness at 0.6, 2.4 and 9.2 km.	TES, USGS SIM 3292, Robbins & Hynek, MOLA
Water and ice	SWIM ice consistency at 0-1 m, 1-5 m and deeper; the nearest ice-consistent ground and its share of the 100 km around; GRS water-equivalent hydrogen.	SWIM, Mars Odyssey GRS
Composition	Plagioclase, pyroxene and glass / clay (TES); iron, chlorine, potassium, thorium and silicon (GRS); the hydrous and aqueous mineral detections nearby.	TES, GRS, CRISM, OMEGA

Section	What it holds	From
Science here	Every catalogued layer within 100 km with its nearest feature, a Special Region warning, and the seven science objectives scored at the point.	the science atlas (Chapter 6)
Missions and sites nearby	Landed and targeted missions by distance and direction, NASA Exploration Zones and Starship candidates.	mission catalogue, candidates
Best data here	The HiRISE 1 m and CTX 20 m elevation models that cover the point, and the finest resolution a hub built here would get.	USGS STAC catalogue

How the values are made

- **Fast.** Everything is read from data on this computer or its caches in well under a second. Two lookups are remote and run in parallel with a four-second budget: the geologic unit (USGS OGC API) and NASA Mars Trek's colour maps. If one is slower, its line says *reading…* and fills in by itself a few seconds later.
- **Colour maps read back.** The TES mineral and dust-cover maps are published as rainbow images under a hillshade. Hillshade changes brightness, not hue, so the hue at the point gives its place on the scale, shown as very low to very high, never as an abundance. Checked on known places: Syrtis Major reads very high for pyroxene and plagioclase; Arabia Terra, Tharsis and Olympus Mons high for dust; Meridiani smooth; Hellas has no TES data, as published.
- **Liquid water.** Above water's triple point (611.7 Pa) pure liquid water can exist between 0 °C and its boiling point at the local pressure (Buck's equation); below it, ice turns straight to vapour.
- **Gravity** from the mass, flattening and spin of Mars at the ground (3.71 m/s^2 at the equator), plus the measured MRO anomaly where the gravity layer is installed.
- **Dose**: Curiosity's RAD record at Gale, moved by the air column overhead (Guo et al. 2017); galactic cosmic rays only, solar particle events come on top.
- **Missing data say so.** A layer that is not installed, or has no value at that latitude, is named as such; nothing is filled in.

> **What the numbers are, and are not**
>
> Every value comes from orbit or a model, at the resolution of its dataset: GRS cells are 5° (about 300 km) wide, the TES maps 3 to 16 km, SWIM 0.05°. They describe the ground around the point, not what a boot or a drill will meet. Pressure and dose are mean, modelled values; colour-map values are relative.

On a built EVA site

In the EVA planner, click any point of the map (when **+ Add on map** and **Target on map** are off). A card opens over the map, named after the nearest station or landmark, with the point's position and its distance and direction from the habitat; a yellow ring marks it on the map. Esc or × closes it.

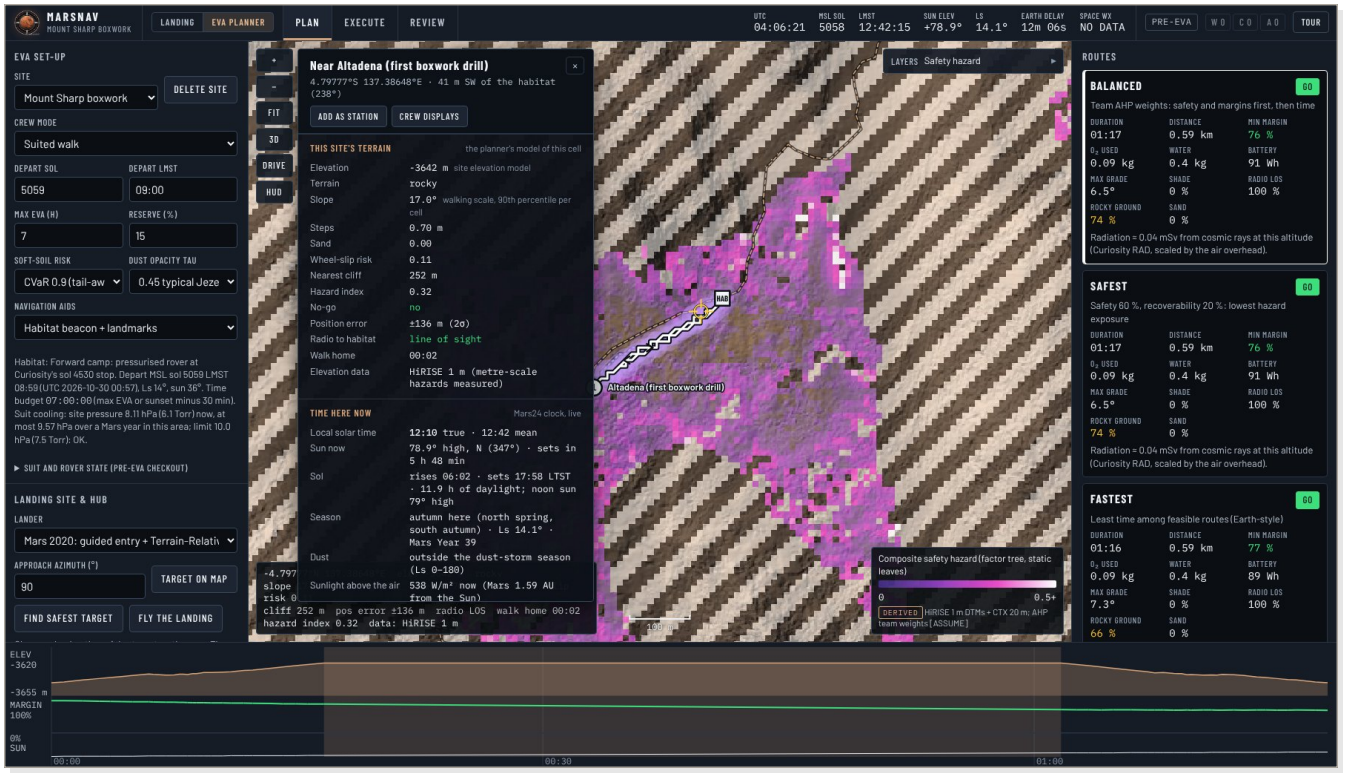


Figure 15. *A point clicked 41 m south-west of the forward camp on the Mount Sharp site, by the Altadena drill site: the planner's own model of the cell (rocky, 17° walking slope, 0.7 m steps, not no-go, line of sight to the habitat, two minutes' walk home, measured at HiRISE 1 m) and the time there; the planet-wide sections follow further down the card.*

Row	What it is
Elevation, terrain	From the site's own elevation model; the terrain class used by the planner.
Slope, steps	Walking-scale slope (90th percentile per cell) and step height.
Sand, wheel-slip risk, rock cover	Bedform evidence and the soft-soil risk a rover would face.
Nearest cliff, hazard index, no-go	The distance to the nearest drop, the composite hazard and, if the cell is forbidden, why.
Position error, radio	The navigation filter's 2σ error there, and line of sight to the habitat.
Walk home	From the walk-home field of the planned EVA (plan one first).
Elevation data	Whether HiRISE 1 m or only CTX 20 m data back the cell.

- **Add as station** makes the point a station (named after the landmark when you clicked one) and re-plans.
- **Crew displays** opens the helmet visor, rover dashboard and wrist cuff as a survey of the point (Chapter 10).
- **Photos** appears when a station with the rover's own photographs is within 150 m (Chapter 8).
- The planet-wide sections need the MarsNav engine. When the EVA planner is served on its own as plain web files, the card keeps the site terrain and says so.

CHAPTER 06 · WHAT TO STUDY, WHERE, AND WHEN IT IS SAFE

Science atlas, objectives, missions and space weather

The challenge asks for at least three kinds of data from several NASA missions, science along the way, and space weather. MarsNav adds a whole-planet science atlas grouped by the kind of measurement, seven science objectives that rank every site, a catalogue of the missions past, present and future, and an EVA advisory from NASA's space-weather service.

The science atlas

Open **Map layers** in Landing mode: under the base imagery and the science layers is **Science atlas · whole planet**, nine groups that fold open. Tick any layer to draw it on the globe; the legend says what the colours mean and cites the source, and the number beside a catalogue is how many features it holds.

Group	Layers	Source
Spectral: minerals	Aqueous minerals (4,572 sites), hydrous mineral detections (1,648), chloride salt deposits (642); plagioclase, high-calcium pyroxene and sheet silicates / high-Si glass	Ehlmann & Edwards 2014; Carter et al. 2013 (CRISM, OMEGA); Osterloo et al. 2010; Bandfield 2002 (TES)
Chemical: elements	Water-equivalent hydrogen, iron, chlorine, potassium, thorium, silicon	Mars Odyssey GRS (Boynton et al. 2007)
Water and ice	SWIM ice; glaciers and glacial valleys (1,847); gullies (4,978); valley networks (59,394 segments); alluvial fans (44)	Morgan et al.; Souness, Brough, Fassett; Harrison 2015; Hynek 2010; Moore & Howard 2005
Thermal and surface	Thermal inertia, albedo, dust cover index, dune fields (547)	Putzig & Mellon 2007; TES; Ruff & Christensen 2002; Hayward et al. 2007
Topographic and geologic	MOLA surface roughness, the geologic map, craters ≥ 1 km	Kreslavsky & Head 2000; Tanaka et al. 2014; Robbins & Hynek
Geophysical	Gravity anomaly	MRO GMM-3 (Genova et al. 2016)
Atmospheric, radiation, power	Surface air pressure, cosmic-ray dose rate, sunlight for solar power	MOLA elevation; Curiosity RAD (Hassler 2014, Guo 2017)
Present-day activity	508 sites: recurring slope lineae, new and icy impacts, active gullies and dunes; candidate Special Regions	Daubar 2013, McEwen 2014, Ojha 2014, Dundas 2014
Missions on the ground	Rover traverses; 66 landers, rovers and science stops	NASA Mars Trek, JPL waypoint feeds



Figure 16. *The science atlas over Nili Fossae and Jezero (about 1,800 km across): aqueous-mineral sites and CRISM / OMEGA hydrous detections, with the legend and its citations. The groups fold open in the Map layers panel.*

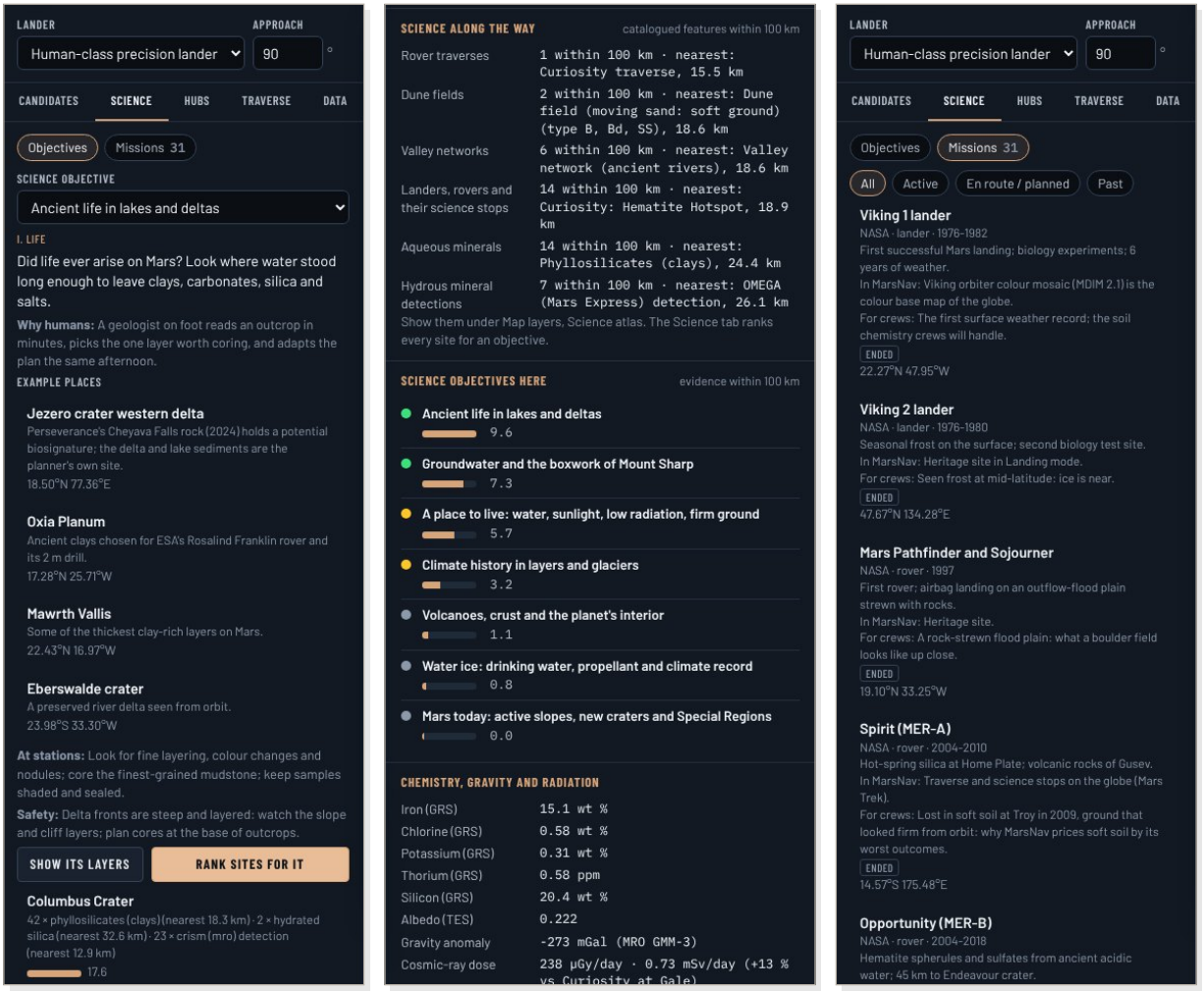
The feature catalogues ship inside the app, built from NASA Mars Trek by `tools/build_features.py`, so they work offline at once. The GRS elements, albedo and gravity install once in the Data tab. The TES mineral, dust and roughness maps stream from NASA Mars Trek and are kept by Local mode. The atmosphere, dose and sunlight maps are computed from MOLA elevation and latitude.

Science objectives

The **Science** tab starts from the question, as the challenge does: choose an objective, then find where to pursue it. Each objective card says what to look for, why a human on foot changes the answer, example places, what to do at the station and what to watch for. **Show its layers** turns on the atlas layers that answer it. **Rank sites for it** scores every candidate and every hub you have built by the evidence within 100 km, each kind counted on a logarithmic scale and weighted, plus terms from the installed layers such as ice, water, dose or sunlight. Click a result to fly there and open it.

Objective	Evidence that scores
Ancient life in lakes and deltas	Aqueous and hydrous minerals, valley networks, fans, chlorides, the geologic map
Groundwater and the boxwork of Mount Sharp	Aqueous and hydrous minerals, rover traverses, dunes, the gravity anomaly (the challenge's example)
Water ice: drinking water, propellant and climate record	SWIM ice, GRS water, glaciers, active slopes, thermal inertia

Objective	Evidence that scores
Mars today: active slopes, new craters and Special Regions	Present-day activity, gullies, dunes
Volcanoes, crust and the planet's interior	Pyroxene, plagioclase, GRS potassium and thorium, gravity, geology
Climate history in layers and glaciers	Glaciers, valley networks, fans, geology, ice
A place to live: water, sunlight, low radiation, firm ground	Ice, GRS water, cosmic-ray dose, sunlight, dust cover, thermal inertia



An objective and its ranking | Science in the dossier | The mission catalogue

Figure 17. *Ancient life ranked over every site (Columbus crater first, with the evidence that put it there); the science sections of the dossier at Mount Sharp; and the mission catalogue, from Viking to the missions still on their way.*

Missions, past, present and future

Missions 31 lists the Mars missions as of 28 October 2026: landers, rovers, orbiters and helicopters that have ended, are active, are en route, are planned or in development, or were cancelled. Filter by **Active**, **En route / planned** or **Past**. Each entry gives the agency, the years, the status (with a source wherever it can change), what the mission found, what it gives MarsNav's map and what it means for a crew; landing sites are on the globe.

Statuses that change quickly (MAVEN lost in 2025, ESCAPADE and MMX en route, Mars Sample Return cancelled in its current form) are dated and sourced in `planet/data/missions.json`.

Space weather

SPACE WX in Landing mode's top bar, and the matching box in the EVA planner's top bar, read NASA's DONKI service (the CCMC Space Weather Database of Notifications, Knowledge, Information): CME arrivals predicted at Mars by the WSA-ENLIL model, solar particle events and flares over the last 30 days, refreshed at most every three hours and kept for Mars mode. The digest also gives the Sun-Earth-Mars angle: past about 60° Mars faces solar regions that Earth-side monitors cannot see, so DONKI may miss events that reach Mars.

Level	When	Advice for EVAs
QUIET	Nothing below applies.	EVA as planned.
WATCH	A CME is predicted at Mars within 72 h; or a particle event near Earth in the last 48 h with Mars more than 60° away; or an X-class flare in the last 24 h.	Shorter EVAs within reach of shelter; recheck before egress.
STORM	A CME shock at Mars within ±24 h; or a particle event in the last 48 h with Mars within 60° of Earth (probably the same magnetic connection).	Hold EVAs or return to shelter (habitat or pressurised rover).

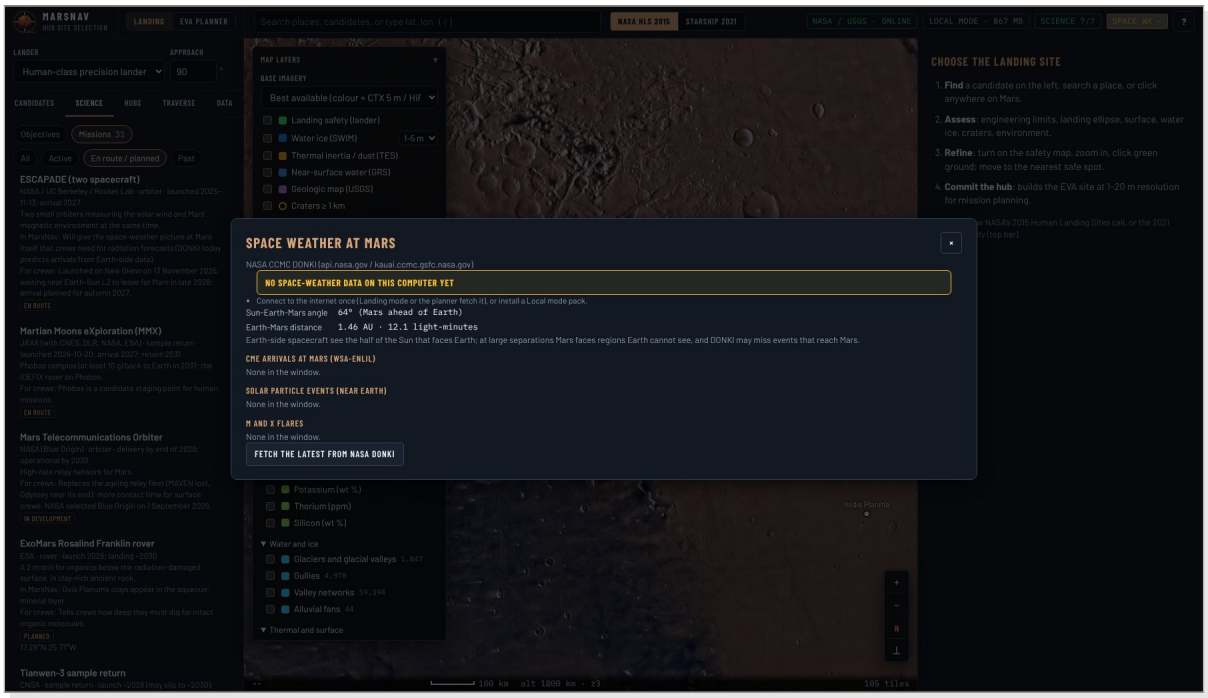


Figure 18. *The space-weather digest. On the day of these screenshots NASA's service refused requests from the development network (rate limit), so the panel says there is no data instead of guessing; the Sun-Earth-Mars geometry is computed locally (64°, 1.46 AU, 12.1 light-minutes).*

Every route card also gives the route's cosmic-ray dose, from the altitude of every step (more air overhead, less dose), and Execute's **Solar particle event** drill rehearses the return. Your own NASA API key, in the environment variable `MM_NASA_API_KEY`, avoids the demonstration key's limit.

CHAPTER 07 · WORKING WITH NO EARTH LINK

Local mode and Mars mode

On Mars there is no internet. Local mode puts every dataset MarsNav uses on this computer, at the resolution it was published at. Mars mode then cuts the app off from the internet, so you can show that everything still works, and the panel counts every request it refused.

Open it from the **LOCAL MODE** pill in Landing mode's top bar, the **Local mode** button in the Data tab, or **Data** menu, **Local mode**.

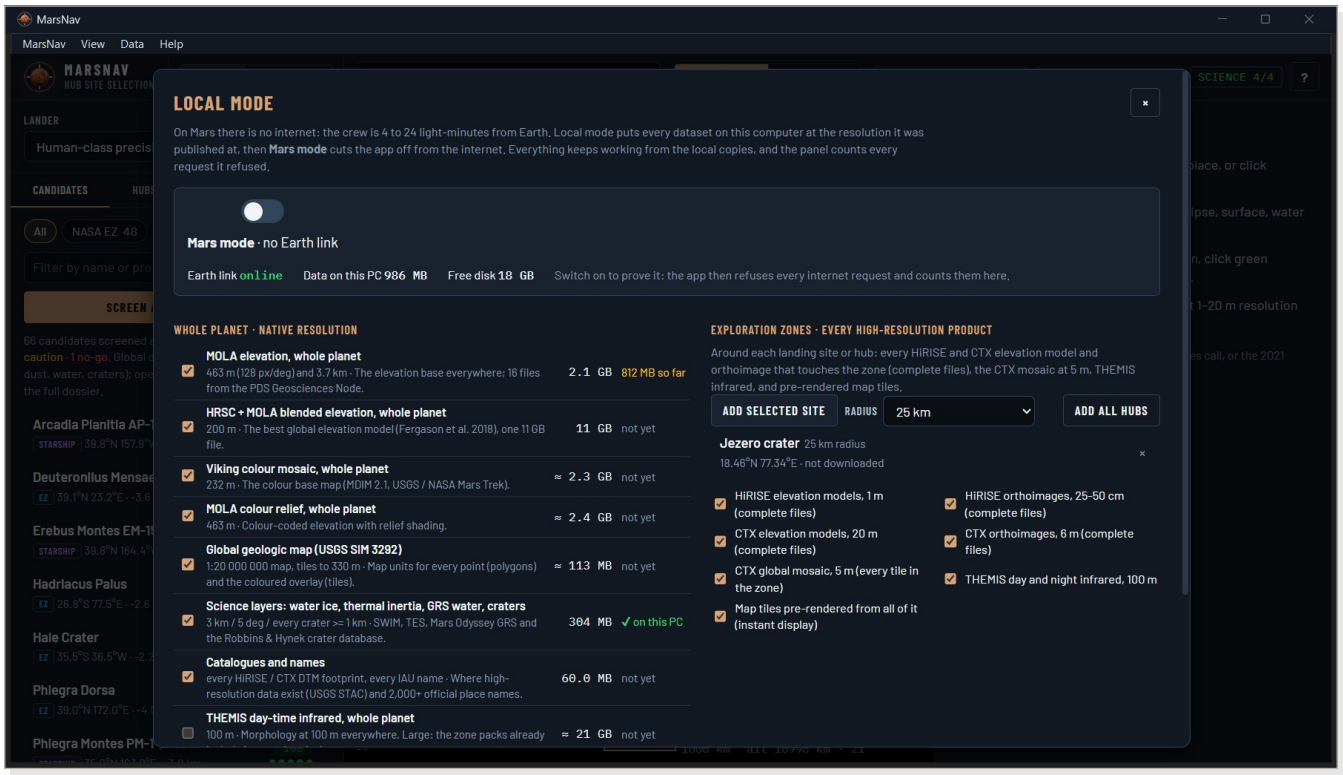


Figure 19. *The Local mode panel after* ***Measure sizes****: whole-planet packs on the left, exploration zones on the right, and the size check against the free disk at the bottom (here: 29 GB selected, 20 GB free, so MarsNav says it will not fit and suggests what to untick, before anything is downloaded).*

What can be downloaded

Whole-planet pack	Resolution	Size
MOLA elevation	463 m (128 px/deg) and 3.7 km; 16 files from the PDS Geosciences Node	2.1 GB
HRSC + MOLA blended elevation	200 m; the best global elevation model (Fergason et al. 2018), one file	11 GB
Viking colour mosaic	232 m (MDIM 2.1, USGS / NASA Mars Trek)	≈ 2.3 GB
MOLA colour relief	463 m, colour-coded elevation with shading	≈ 1.7 GB

Whole-planet pack	Resolution	Size
Global geologic map (USGS SIM 3292)	unit polygons for every point, and the coloured overlay	≈ 113 MB
Science layers	SWIM ice, TES thermal inertia and albedo, GRS water and elements, MRO gravity, every crater ≥ 1 km	≈ 310 MB
Mineral and surface maps	TES plagioclase, pyroxene, sheet silicates and dust cover; MOLA roughness (NASA Mars Trek)	≈ 6 MB
Catalogues and names	every HiRISE / CTX elevation-model footprint, 2,000+ IAU names	60 MB
THEMIS day / night infrared (optional)	100 m, whole planet	≈ 21 GB each

Exploration zones add full resolution where the crew will actually work. Add the site selected on the globe (or all your hubs) with a radius of 10 to 50 km, and choose what to include: every HiRISE elevation model (1 m) and orthoimage (25-50 cm) that touches the zone as complete files, every CTX elevation model (20 m) and orthoimage (6 m), every tile of the CTX 5 m global mosaic in the zone, THEMIS infrared, and map tiles pre-rendered from all of it.

> **Why not simply 'download all of Mars'?**
> The complete HiRISE and CTX archives are hundreds of terabytes. MarsNav therefore keeps every whole-planet dataset at its native resolution and the full-resolution archives inside the zones you choose, which is what a crew on the surface needs. Sizes are measured against the real files before anything is downloaded.

Downloading

- **Measure sizes** asks the NASA and USGS servers for the exact size of every file and compares the total with the free disk (keeping 2 GB free).
- **Download to this computer** shows overall and per-item progress, speed and time left. Downloads run in their own process, resume where they stopped, and can be stopped with **Stop**.
- Everything goes to `data_cache\planet\local` with a manifest and checksums; tiles you have already viewed are reused.

Mars mode

Switch on **Mars mode** in the panel. From that moment the engine refuses every outside request instantly and counts it; the top bar turns orange and reads **EARTH LINK OFF** and **MARS MODE · LOCAL DATA ONLY**. Maps, assessments, the EVA planner and the closed loop keep working from the local copies. The interface itself, fonts included, is part of the app, so it looks the same with no internet. Switch Mars mode off and MarsNav may use the internet again.

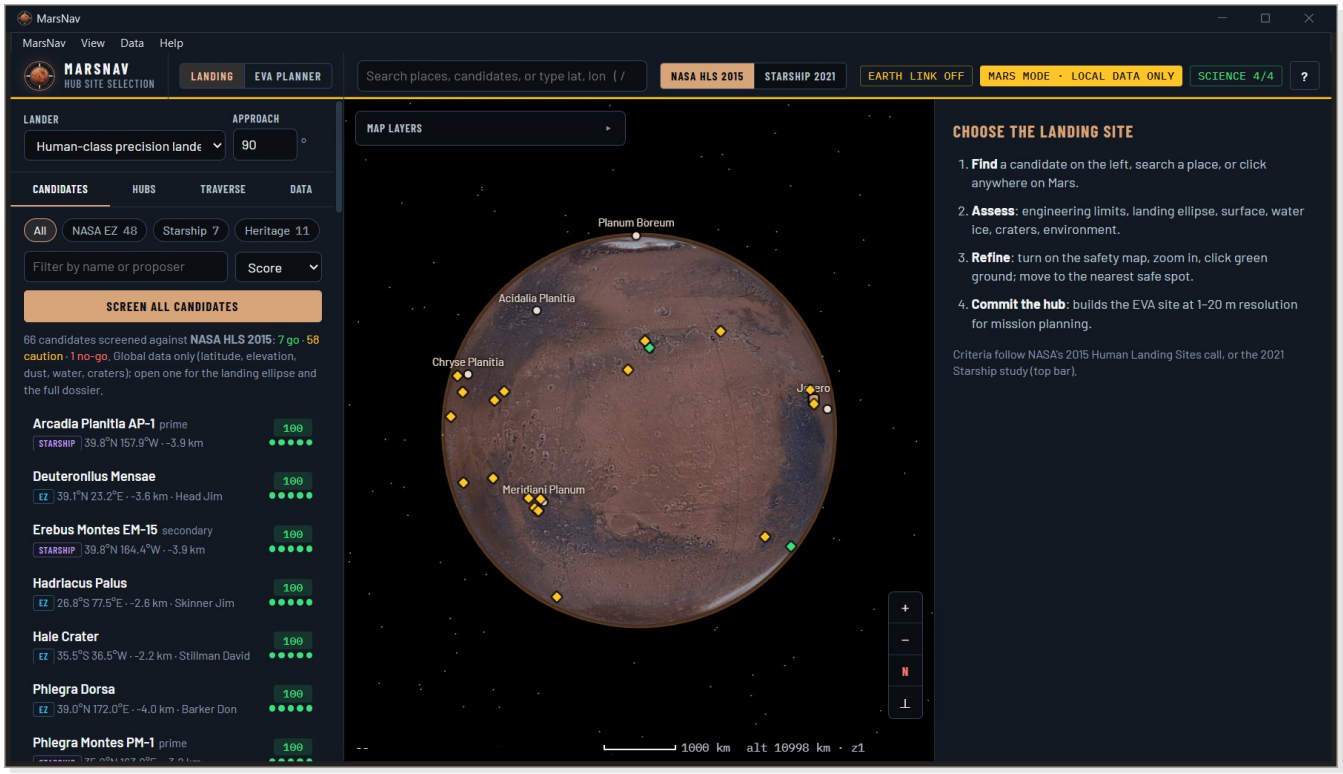


Figure 20. *Mars mode on: the whole globe drawn from data on this computer, with the orange EARTH LINK OFF / MARS MODE banner in the top bar.*

Demonstrating it to judges

- Open Local mode and switch on **Mars mode**.
- Fly around the globe, assess a candidate, open the EVA planner and plan an EVA.
- Open the panel again: **Requests refused** counts the outside requests the app refused while it kept working from local data.
- For a fully convincing demo, also unplug the network: nothing changes.

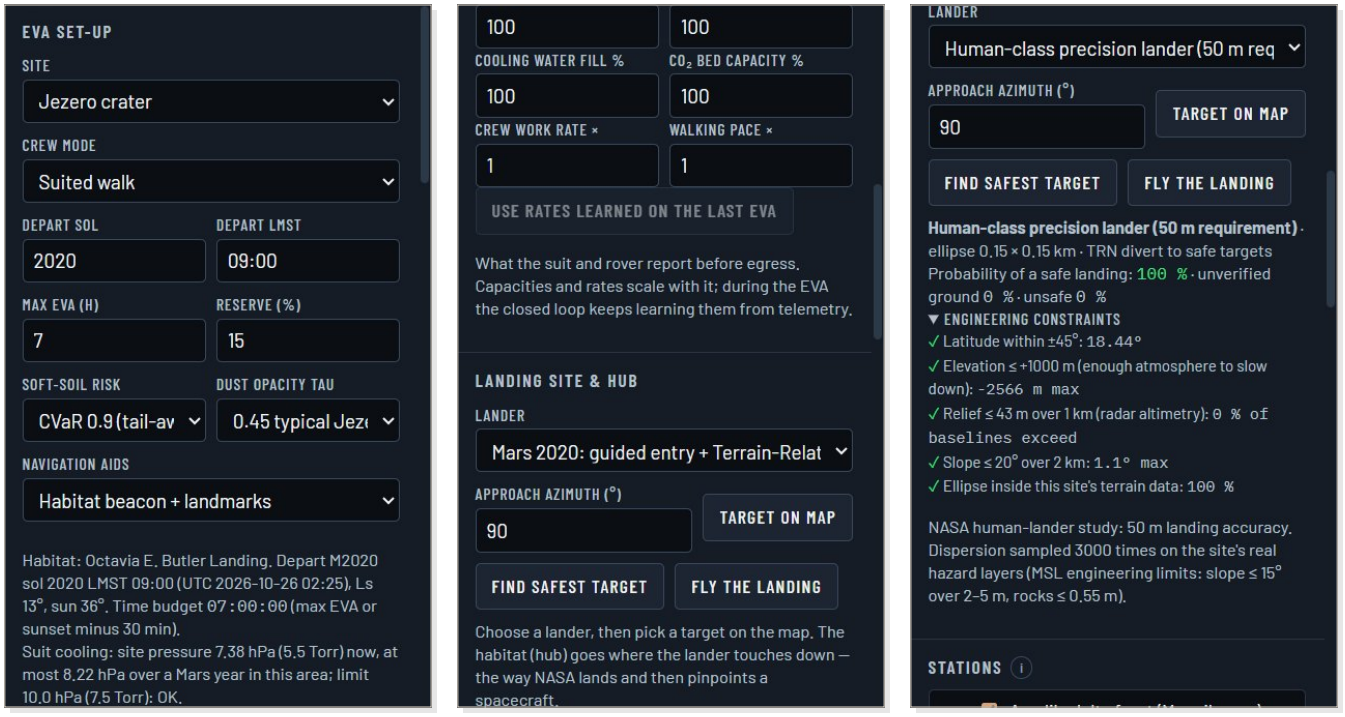
CHAPTER 08 · THE EVENING BEFORE THE MARSWALK

EVA planner: Plan

Plan is where the crew and science team decide tomorrow's EVA: where to go, what to do there, and whether everyone can always get home. It opens by default and runs entirely on this computer.

The demonstration site is Jezero crater, which has the richest public data on Mars: 1 m HiRISE stereo terrain, a 45 km rover traverse and years of MEDA weather records. The planner itself is site-agnostic (Chapter 13).

Set up the EVA



EVA set-up | Suit and rover checkout | Landing site & hub

Figure 21. *The left panel in Plan mode: the EVA conditions, the pre-EVA checkout state of suit and rover, and NASA's landing logic applied at site scale.*

- **Crew mode**: suited walk, unpressurised rover with walkback, or robotic scout (for validation against rover data).
- **Depart sol / Depart LMST**: when the EVA starts, in Mars local mean solar time. **Max EVA** and **Reserve**: the time budget and the margin held back from every consumable.
- **Soft-soil risk** (how cautious to be about ground that looked firm, priced by its worst cases) and **dust opacity** .
- **Navigation aids**: habitat beacon and landmarks, dead reckoning and landmarks only, or a studied orbital navigation constellation. The position error they imply widens the stand-off from cliffs and keep-out zones.

- **Suit and rover state**: tank fills, battery health, scrubber capacity, the crew's work rate and pace, rover battery and wheel wear. **Use rates learned on the last EVA** starts from what the closed loop learned.
- Below the fields: the site's suit-cooling check (pressure against the 7.5 Torr evaporative-cooling limit) and the EVA time budget (the EVA limit, or sunset minus 30 minutes).

Landing site & hub at site scale

Choose a lander (Viking, MER, MSL, Mars 2020 with Terrain-Relative Navigation, or a human-class precision lander) and an approach azimuth. **Target on map**, **Find safest target** and **Fly the landing** run a Monte Carlo of 3,000 touchdowns over the lander's delivery ellipse on the site's 1 m and 10 m hazard maps; with TRN the lander diverts to safe ground as Perseverance did. The result is the probability of a safe landing and the engineering constraints met. **Set as hub** moves the habitat, the start and end of every route, to the touchdown point.



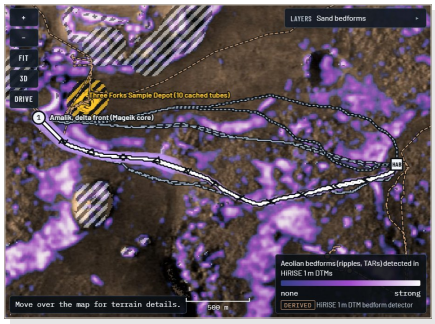
Figure 22. *The landing-safety layer at Jezero for a human-class lander: the crater floor is safe (green), steep ground hazardous (red). Probability of a safe landing at the chosen target: 100 %.*

Read the terrain: eleven layers

The **Layers** box switches what the map shows. Every layer is computed from real data on a 10 m grid (2.3 million cells at Jezero), and its legend names the source and whether it is derived, modelled or an assumption.



Safety hazard (composite)



Sand bedforms



Walkback time

Figure 23. *Composite safety hazard; aeolian bedforms detected in the 1 m DTMs (the ripples that trapped Spirit and Opportunity); and walk-home time to the habitat with 15-minute contours, the field the walk-home guarantee rests on.*



Slope (walking scale)



Shadows at time

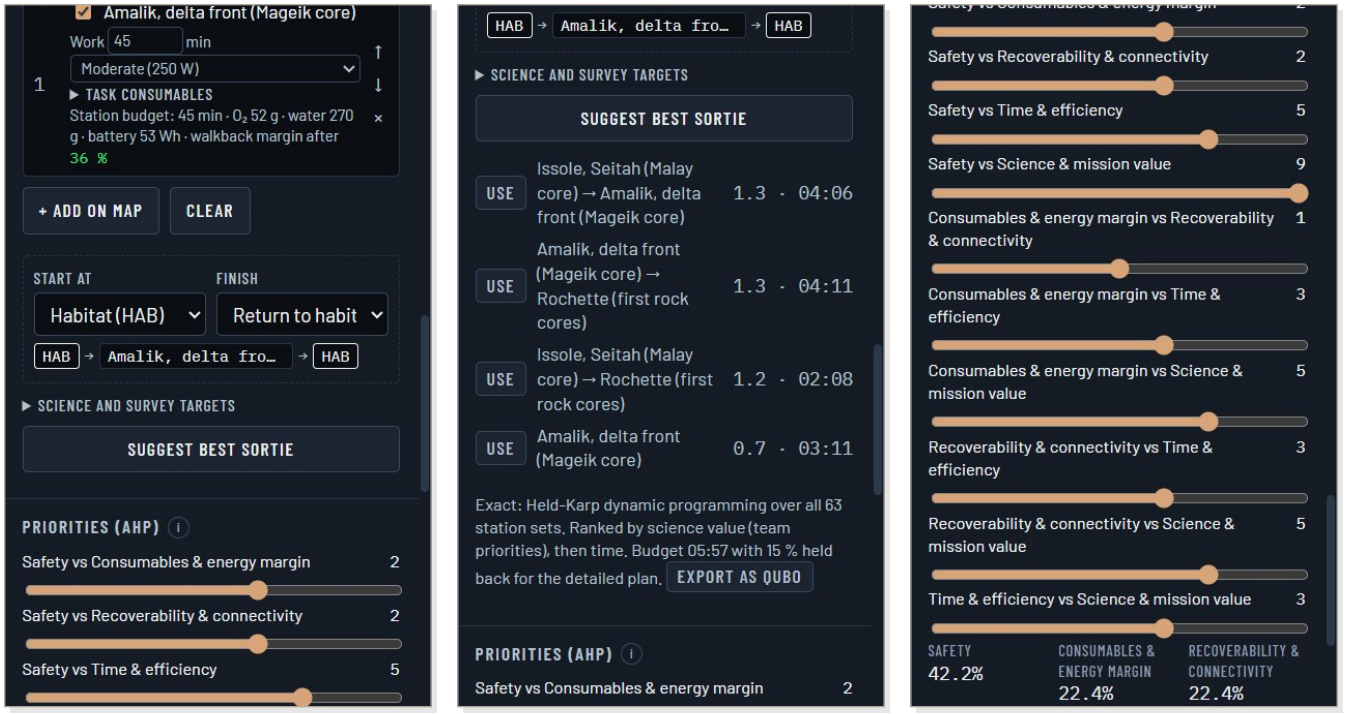


Radio line of sight

Figure 24. *Walking-scale slope (the 90th percentile in each 10 m cell, from 1 m data); terrain shadow at the selected time from horizon maps and the Mars24 sun; and where the crew loses radio line of sight to the habitat mast.*

Layer	What it shows
Safety hazard	A composite risk score from a factor tree (slope, steps, roughness, sand, soft soil, cliffs).
Slope · Sand bedforms · Wheel-slip risk	The ground a boot or wheel meets.
Shadows at time · Walkback time · EVA range	Light, the walk home, and how far the consumables reach.
Radio line of sight · Position error · Data confidence	Communications, navigation uncertainty, and where the data are weak.
Landing safety	Lander-scale safety for the chosen lander.
No-go areas · Perseverance traverse · Landmarks · Alternative routes	Overlays: cliffs and keep-outs (hatched), the rover's real path, navigation landmarks, the other route options.

Choose stations and the best sortie



Stations and route order | Suggest best sortie | Priorities (AHP)

Figure 25. *Stations with their tasks and consumables, the exact sortie optimiser, and the AHP priorities with a live consistency ratio (CR 0.009: consistent).*

- Add stations with **+ Add on map**, or pick from the curated **science and survey targets** (including Perseverance's real sample sites). For each, set the work time, intensity (light, moderate or heavy), tool energy and any extra oxygen or water; its own budget and the walk-home margin after it update at once.
- Tick the stations to visit, order them with ↑ / ↓, and choose where the route **starts** and **finishes** (back at the habitat, at the last station, or at any point, for example a rover pick-up). The walk-home guarantee holds whatever the end point.
- **Suggest best sortie** finds the best set and order of stations for the time, oxygen and battery budget, exactly (Held-Karp dynamic programming over every subset, about 35 ms for 15 candidates), ranked by science value.
- **Priorities (AHP)**: weigh safety, consumables, recoverability, time and science value with pairwise sliders. The consistency ratio updates live, so you can see when your own judgements stop being consistent.
- **What you'll see · n photos** on a station opens the rover's own photographs from that spot (below).
- Click any point of the map, outside **+ Add on map**, for the site's terrain there and every planet-wide layer; **Add as station** on that card adds it (Chapter 5).

Photographs from the ground

The challenge asks for point-of-view images so the crew knows what to expect. Where a rover has stood at a station, MarsNav keeps that rover's own raw Navcam and Mastcam frames from the spot: four at each of the eight Jezero stations (Perseverance) and two at each of the three Mount Sharp drill sites (Curiosity). Perseverance frames carry the rover's site and drive index, so only frames taken at that exact stop are used; for Curiosity, only

sols when it stood there all day. Sun images, sky frames, deck shots and near-duplicates are rejected. Every photo carries the sol, local time, pointing, the rover's distance from the station and the credit, and links to NASA's full-resolution original. The same photos appear on the wrist cuff for the next stop (Chapter 10).

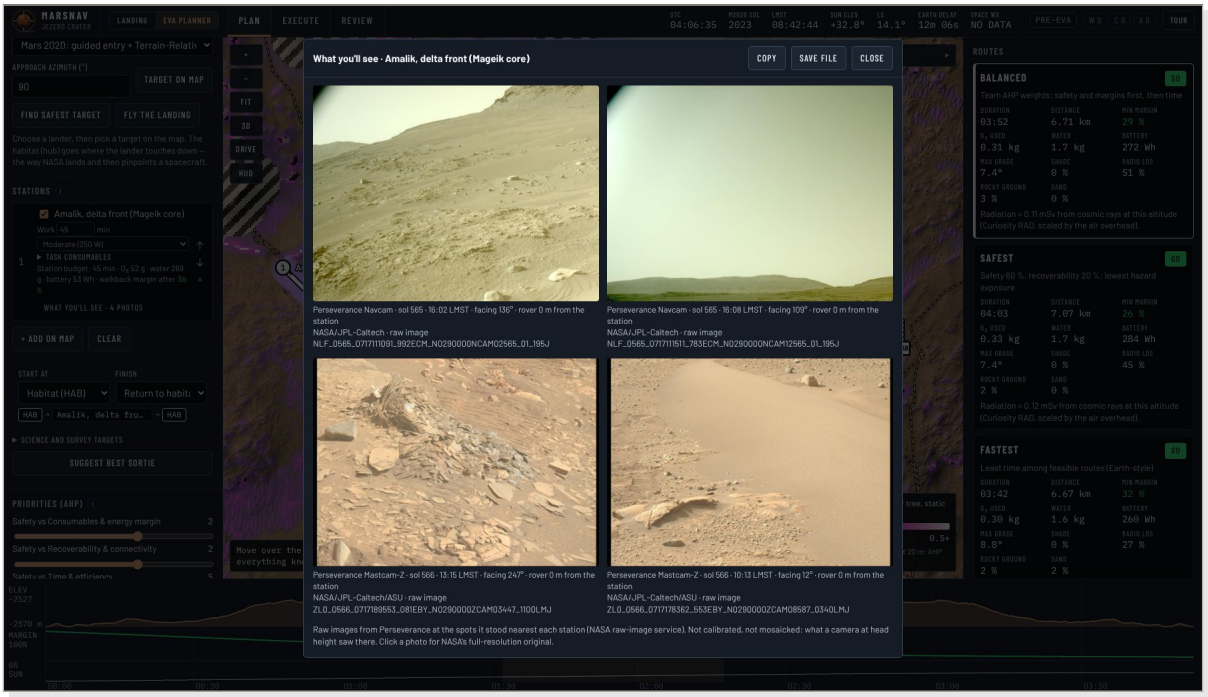
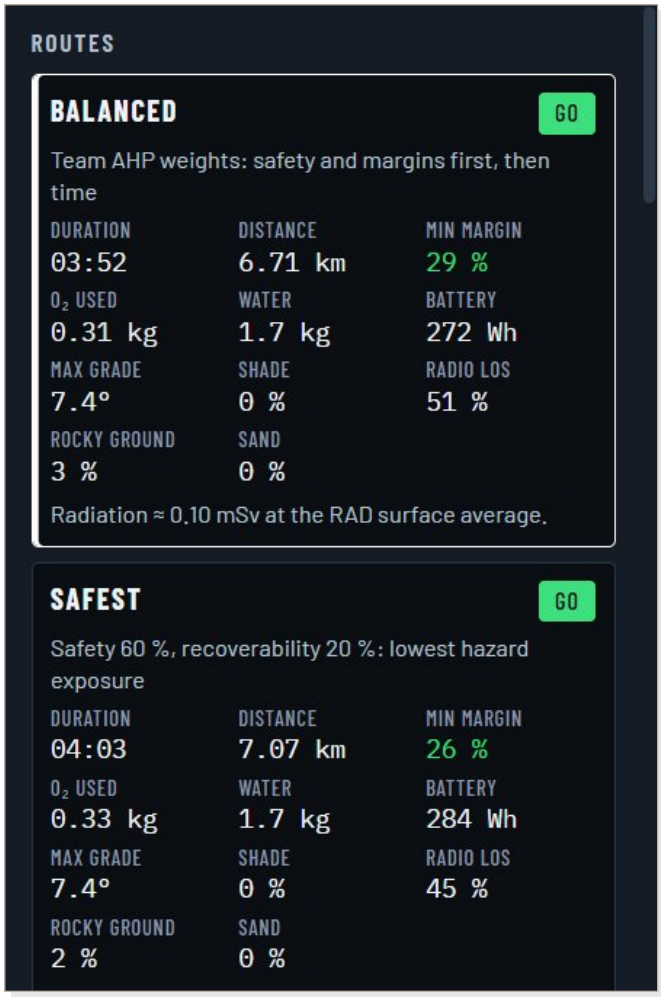


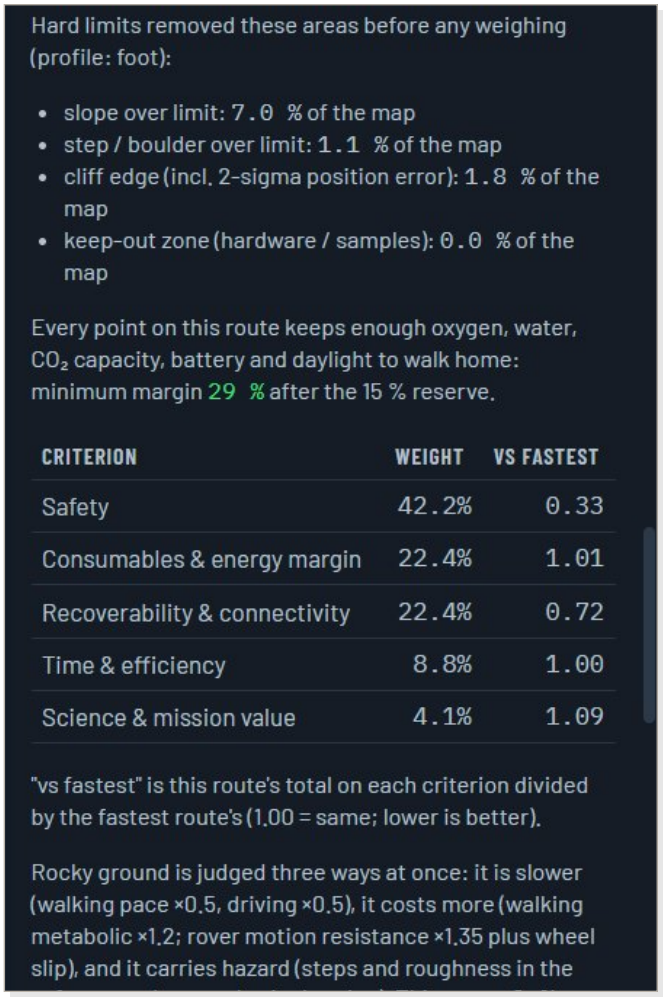
Figure 26. *What you'll see at Amalik, the delta front where Perseverance cored Mageik: two Navcam and two Mastcam-Z frames the rover took from that stop, with sol, time, pointing and credit (NASA/JPL-Caltech/ASU).*

Plan the EVA: five named routes

Press **Plan EVA**. In about three seconds on one laptop core the engine returns five routes, each inside every hard limit: **Balanced** (the team's AHP weights), **Safest**, **Fastest**, **Lowest consumption**, and a **Fallback** in a different corridor, kept ready if the primary is blocked.



Route cards



Why this route

Figure 27. *Each route card shows duration, distance, minimum walk-home margin, oxygen, water, battery, maximum grade, shade, radio line of sight, rocky ground, sand and radiation dose.* ***Why this route*** *lists the hard limits applied and how each criterion compares with the fastest route.*

The dose on each card follows the altitude of every step: Curiosity's measured dose at Gale, scaled by the air overhead, so a route over a ridge costs a little more than one along the crater floor. The space-weather box in the top bar says whether NASA's DONKI service sees a reason to shorten EVAs (Chapter 6).



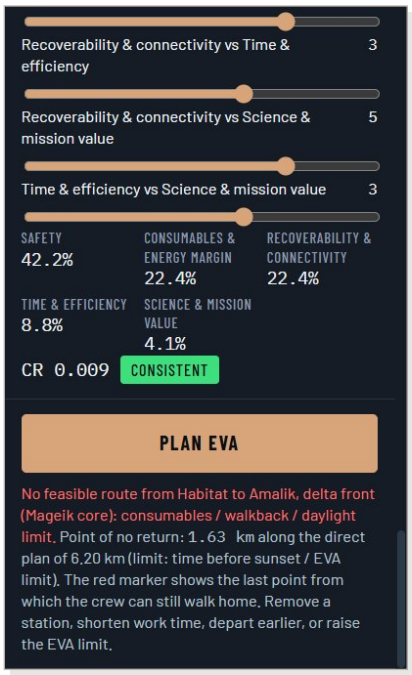
Figure 28. *The timeline under the map: elevation, walk-home margin (green) and sun elevation along the whole route; the shaded band is station work.*

When no plan can work: the point of no return

If no route satisfies every hard limit, MarsNav does not relax a safety rule or hide the problem. It marks the exact **point of no return** on the map, the last point from which the crew can still walk home, and names the resource that runs out first (oxygen, water, battery, CO_2 scrubber or daylight), with what to change.



Point of no return on the map



What to change

Figure 29. *An EVA limit cut to 1.5 hours: no feasible route to the delta front. The point of no return is 1.63 km along the 6.2 km direct plan, and the limit is time before sunset / the EVA limit.*

Rehearse it in 3D

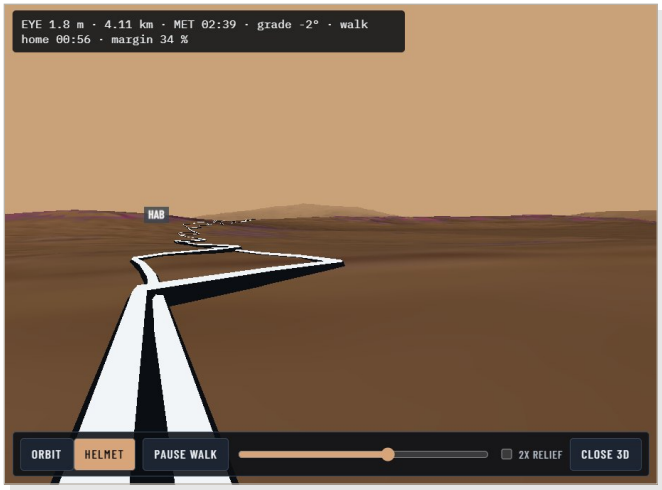
Press **3D** to see the route on the real terrain. **Orbit** above it, switch to **Helmet** for the view at eye height, and press **Walk route** to walk it with the walk-home margin and grade shown at every step. The crater rim is on the real skyline. Esc or **Close 3D** returns to the map.



Figure 30. *3D orbit view of the planned route over Jezero's real terrain (10 m elevation, 2 m HiRISE relief).*



Helmet view at the start



Walking the route

Figure 31. *Eye-height rehearsal: the route ribbon ahead, distance, mission elapsed time, grade, walk-home time and margin in the readout at the top.*

CHAPTER 09 · FLYING THE MARSWALK

EVA planner: Execute

Execute is the suit and habitat display during the EVA. It flies the chosen route in real time (or faster for a demonstration), reads suit telemetry, keeps the walk-home margin in view, raises cautions and warnings, and re-plans from where the crew really is.



Figure 32. *Execute mode at the first station, at 120x speed: consumables (left), the glance band, the live map, alerts and the closed loop (right).*

#	Part	What it shows
1	Consumables	Oxygen, CO_2 scrubber, cooling water and battery used; the white tick is what the walk home would need from here.
2	Glance band	Mission elapsed time, walk-home time, position error (2σ) and radio line of sight.
3	Walkback margin	The worst margin along the rest of the route, and oxygen left.
4	Mode	TRAVERSE, STATION or RETURN.
5	Telemetry source	Built-in simulator, live suit telemetry, or you in the 3D drive window.
6	Hidden truth	What the crew and hardware are really doing. The software is not told; it has to learn it.
7	Run, Pause, Speed	30x, 120x or 600x, so a four-hour EVA can be shown in minutes.

#	Part	What it shows
8	Annunciators	EVA phase and telemetry source (here EVA · SIM TLM), then the warning, caution and advisory counters.
9	Alerts	Latching cautions and warnings with the action to take; acknowledge, or silence the tone.
10	Closed loop	Each rate as planned, as learned from telemetry (with its uncertainty) and the hidden truth.
11	Live map	The crew position with its error radius, the route flown and any re-planned route.
12	Timeline cursor	Where the crew is along the planned profile.

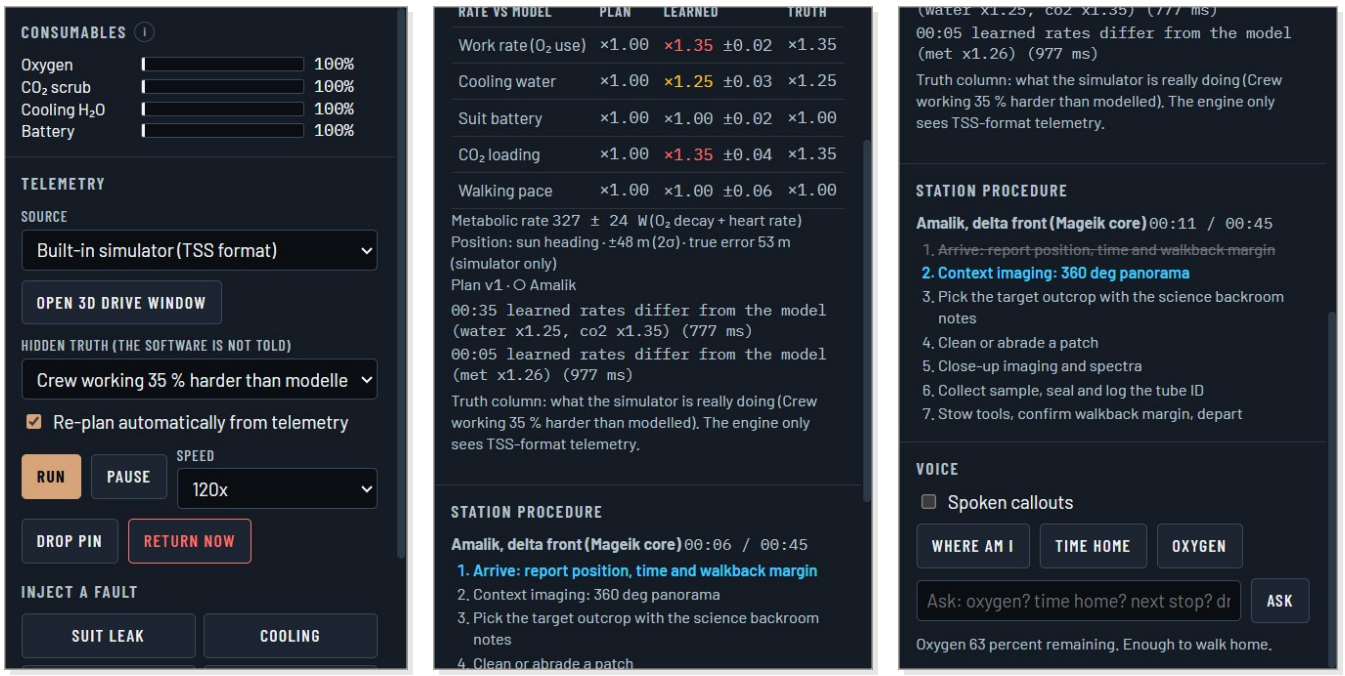
Telemetry sources

- **Built-in simulator (TSS format)**: no setup. Choose a **hidden truth** (crew working 35 % harder than modelled, rough going 25 % slower, a suit battery draining 40 % faster, ground softer than mapped) and watch the closed loop discover it.
- **Live suit telemetry**: **Enable live telemetry** and MarsNav looks for a NASA SUITS Telemetry Stream Server by itself (Chapter 14). With none on the network, a built-in demo suit flies the plan instead.
- **You, in the 3D drive window**: walk or drive the route yourself; your movement becomes the telemetry.

The same EVA can be shown at once on the crew's own displays: **Helmet visor**, **Rover dash** and **Wrist cuff** under the telemetry controls, or **HUD** on the map (Chapter 10).

The closed loop

The heart of Execute. The engine compares the rates the plan assumed with what telemetry reports: work rate (oxygen use), cooling water, suit battery, CO_2 loading and walking pace. As the estimates firm up, it re-plans the rest of the EVA from the crew's real position, shown as a separate **re-planned** route, before the margin is gone. It waits for confident estimates unless the walk-home margin is thin or station work is about to start, and it cuts the lowest-value station short before dropping it.



Telemetry and controls | Closed loop | Station procedure and voice

Figure 33. *The simulated crew is working 35 % harder than modelled. Within minutes the loop has learned the work rate, cooling water and CO_2 loading (×1.35, ×1.25, ×1.35 against the hidden truth of ×1.35, ×1.25, ×1.35), each with its uncertainty. At the station: the procedure checklist and voice queries.*

Caution and warning

MarsNav follows NASA's three-tier convention: **WARNING** (red, an immediate threat such as a suit leak), **CAUTION** (amber, a developing problem) and **ADVISORY** (informational). Every alert has the action to take and a tone; alerts latch until acknowledged, one press silences the tone, and a critical fault turns the plan into the walk-home route automatically. **Return now** is a deliberate two-step action, arm then confirm.

Inject faults with the buttons under the controls: **Suit leak**, **Cooling**, **Injury**, **Dust storm**, **Solar particle event** and **Rover failure**.



Figure 34. *A suit leak injected after the station: SUIT PRESSURE DECAY (warning), and the mode switches to RETURN at once, with 53 % oxygen and a 27 % walk-home margin left; the return route is drawn in red.*

At the station

- The **station procedure** shows the current and completed steps: report position and margin, panorama, pick the target, abrade, close-up imaging and spectra, collect and seal the sample, stow and depart.
- **Drop pin** marks the estimated position with its own error.
- **Voice**: type or speak short questions (Where am I, Time home, Oxygen, Next stop). Replies come only from known telemetry, never invented; tick **Spoken callouts** to hear them.

The 3D drive window

Open 3D drive window (or **Drive** on the map) opens a first-person view of the real terrain in a second window. Choose **Walk** (suited) or **Rover**. W / A / S / D or the arrows move (Shift to hurry), the mouse looks around, F works at a station, C changes the camera, M zooms the minimap and P hides the panel. Every second of your movement becomes suit telemetry in TSS format for the main window, which runs its closed loop on it and re-plans live.

The drive panel sets the hidden truth and the declared checkout, and **Live: change consumables now** hits the running EVA with a sudden event of 15, 35 or 60 %: O_2 tank loss, battery failure, coolant leak, suit O_2 leak, CO_2 scrubber degradation, an exertion spike or a rover battery fault. Each reaction of the planner is listed with how long it took.

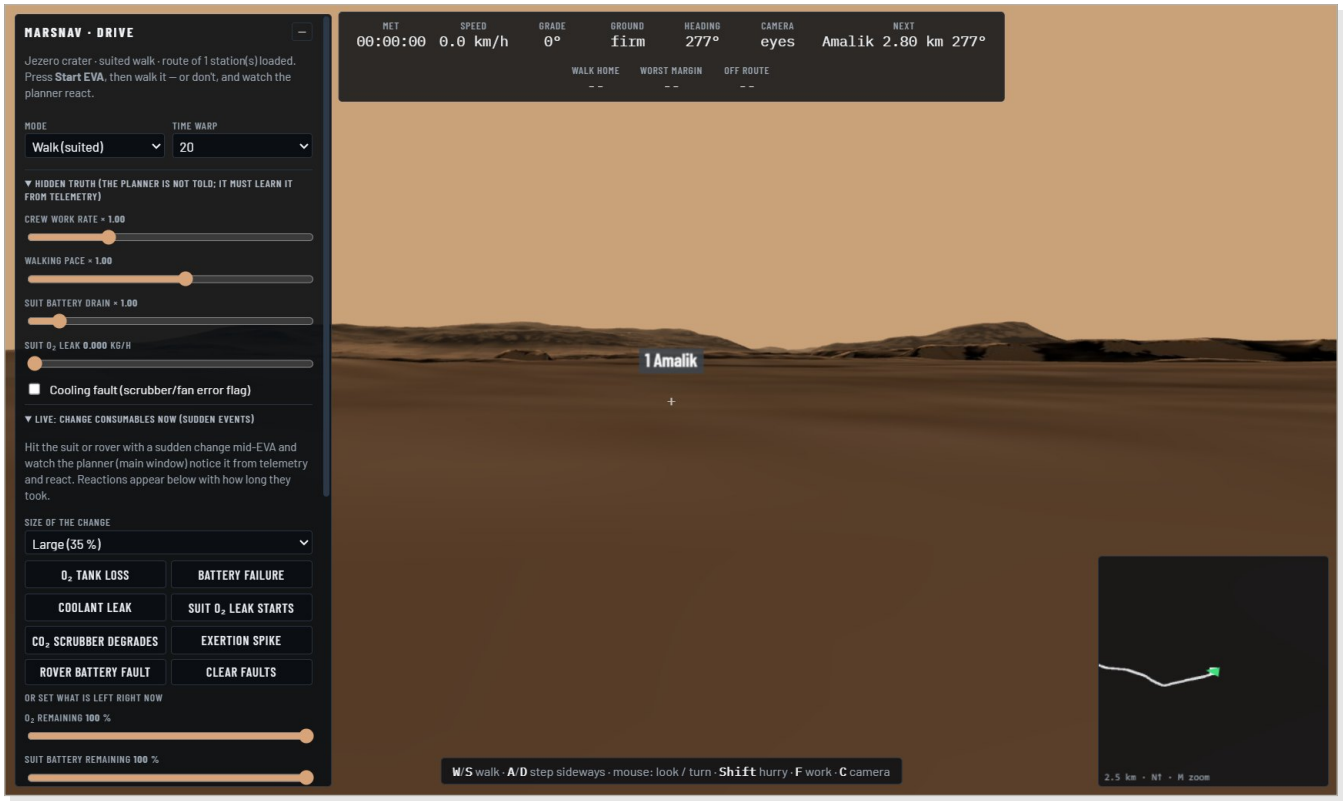


Figure 35. *The 3D drive window: hidden truth, live consumable events and the first-person view of Jezero with the route, the next station and the minimap.*

CHAPTER 10 · HELMET VISOR, ROVER DASHBOARD AND WRIST CUFF

Crew displays

The challenge asks how the guide is shown: inside the habitat, on a tablet, or as a heads-up display on the helmet visor. MarsNav shows the same EVA, live, on the three displays a crew would carry: a visor HUD, the dashboard of the crew rover, and a wrist cuff made for gloved hands. Each shows the map under the crew and the facts about the ground they stand on.

Opening them

- In the EVA planner: **HUD** on the map tools opens the visor; **Helmet visor**, **Rover dash** and **Wrist cuff** in Execute open each one. Every display is its own window: keys **1**, **2** and **3** switch between them and **F** goes full screen.
- They follow whatever flies the EVA: the simulator, live suit telemetry or you in the drive window. They get the plan and every snapshot live, and the cuff's commands act on the planner's loop. A display opened first waits for the planner and connects by itself.
- In the 3D drive window, **H** draws the visor see-through over the first-person view.
- For a place with no EVA, **HUD** in Landing mode's dossier and **Crew displays** on the planner's place card open them as a survey of that point (below).

Helmet visor



Figure 36. *The helmet visor during a simulated EVA to Amalik at 120x: heading 254° with the next stop 1.88 km away (turn right), consumables with the walk-home need ticked, the walk-home time and margin, the facts about the ground under the crew (bottom left) and the mini map.*

Part	What it shows
Heading tape	The 120° ahead with the next stop (diamond), home (H) and the sun. Heading comes from the track: SUITS telemetry has no compass.
Steering cue	An arrow to the next stop with its distance, bearing, time at the current pace and turn left / right / ahead; amber when the next stop is home.
Mode tag	TRAVERSE, STATION, RETURN or EVA DONE.
Consumables	O_2, CO_2 scrubber, cooling water and battery; the white tick is what the walk home from here needs.
Right column	Walk-home time, worst margin, mission time, position error, radio, the sun's height and time to sunset, and the dose so far.
Here	Air pressure, dose rate, gravity, the boiling point of water, the ground, dust, ice and the geologic unit at the crew's position (Chapter 5).
Alert and step	The top unacknowledged alert with its action, and the station procedure step.
Mini map	The route, the re-plan or the return, the track, the stations, the habitat and the crew with its position error; lines only.



Figure 37. *The same visor seconds after a suit leak is injected: WARNING · SUIT PRESSURE DECAY with its action, the mode RETURN, and the steering cue turned amber towards home.*

Why the visor looks like this

Black is see-through on a real visor, so it draws nothing it does not need and keeps the centre of view clear. Normal symbols use one calm colour; amber and red appear only for cautions and warnings. This follows the display practice of NASA's SUITS challenge and NASA-STD-3001, as the planner's own display does.

Rover dashboard

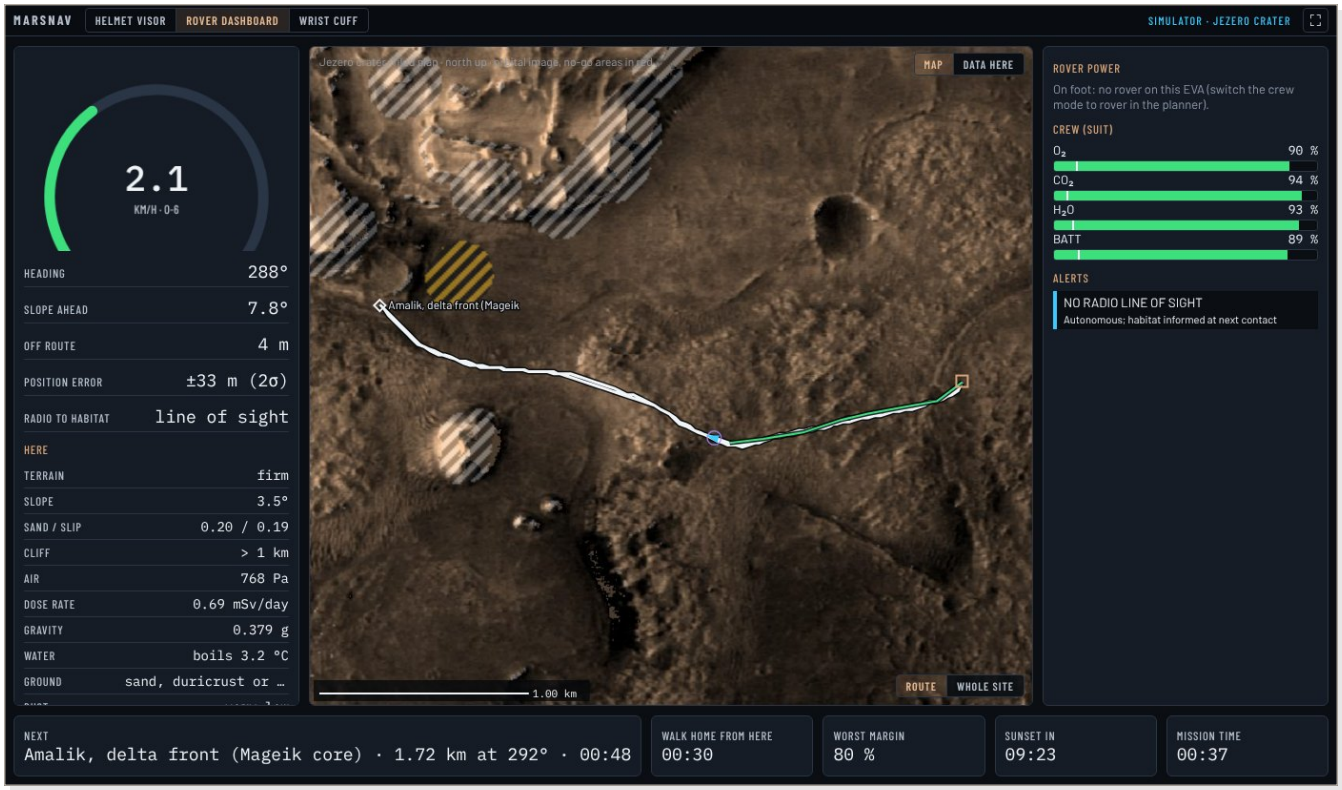
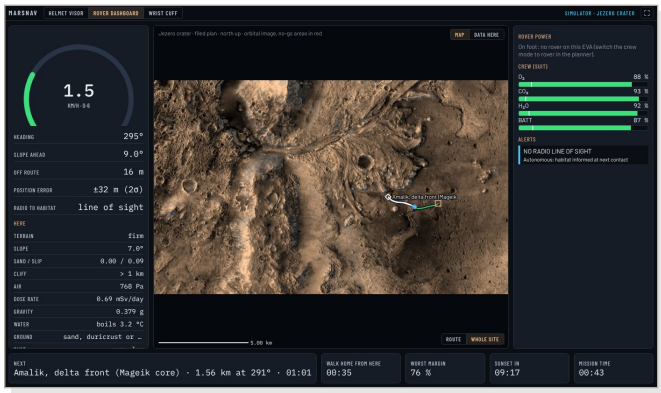
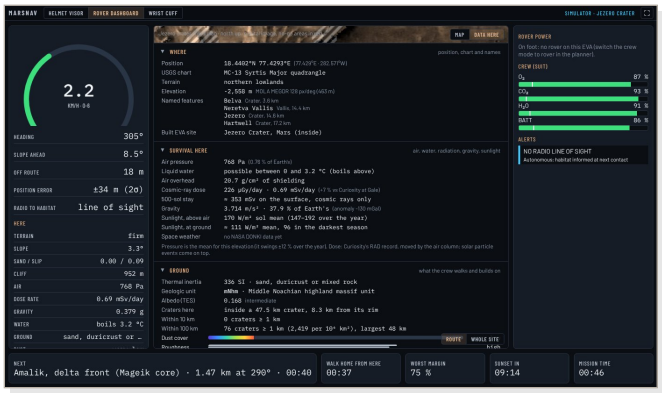


Figure 38. *The rover dashboard on the same EVA: speed, heading, slope ahead, off-route distance, position error and radio (left), with the site's terrain under the crew and the place facts below; the map on the orbital image of Jezero with the no-go areas hatched, the route in white and the track in green (centre); suit consumables and alerts (right).*

- **Map**: drawn on the site's own orbital image. Zoomed in, HiRISE detail tiles replace the 5 m image. **Route** fits the EVA, **Whole site** shows the operations area.
- **Data here**: every layer at the crew's position, the same sections as Chapter 5, folded.
- **Rover power** shows the battery left when a rover is on the EVA; on foot it says so.
- Along the bottom: next stop, walk home, worst margin, sunset and mission time.



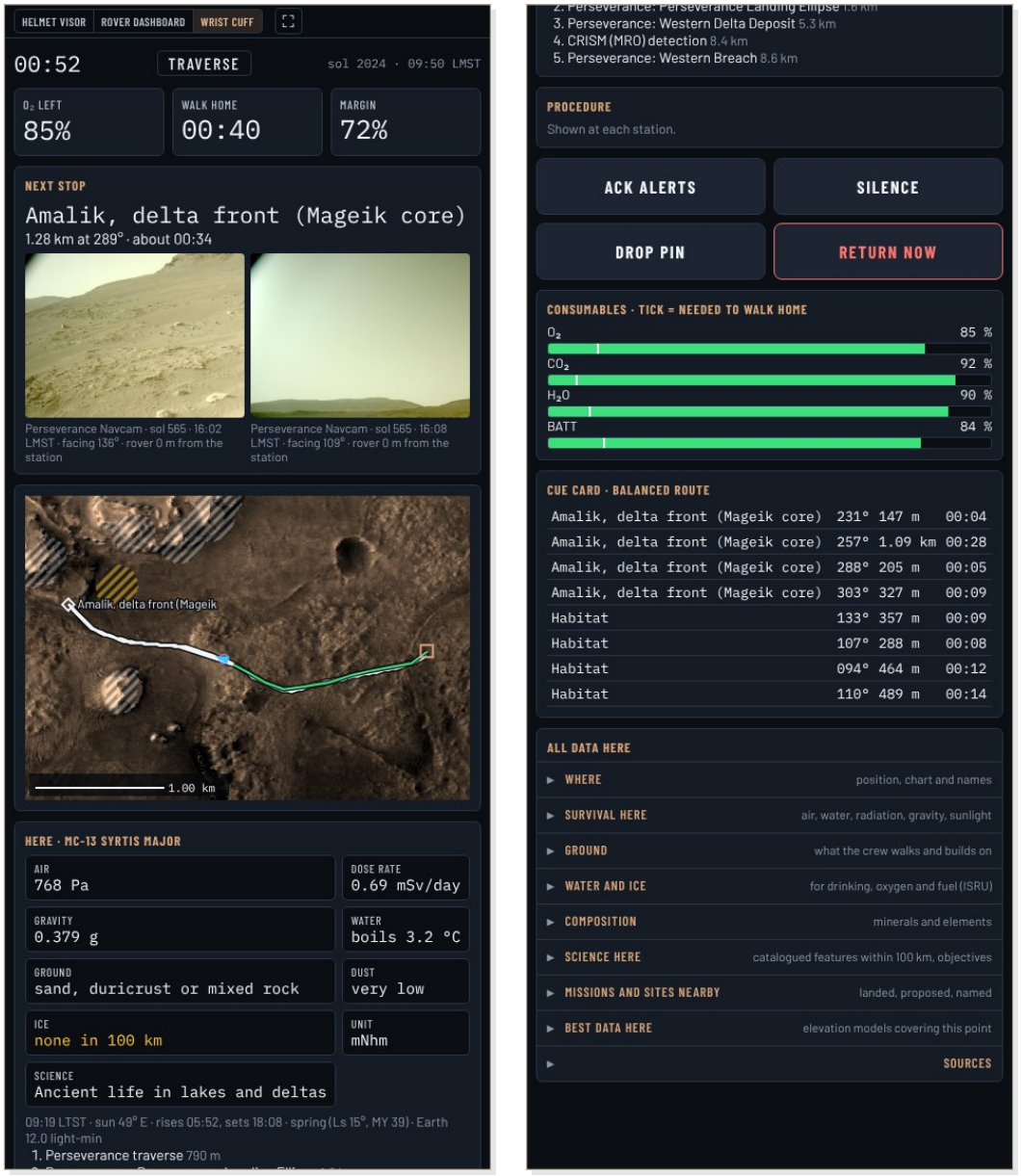
Whole site



Data here

Figure 39. *The dashboard's map over the whole Jezero site, and the Data here tab with every layer at the crew's position.*

Wrist cuff



Top: alert, next stop, map, here

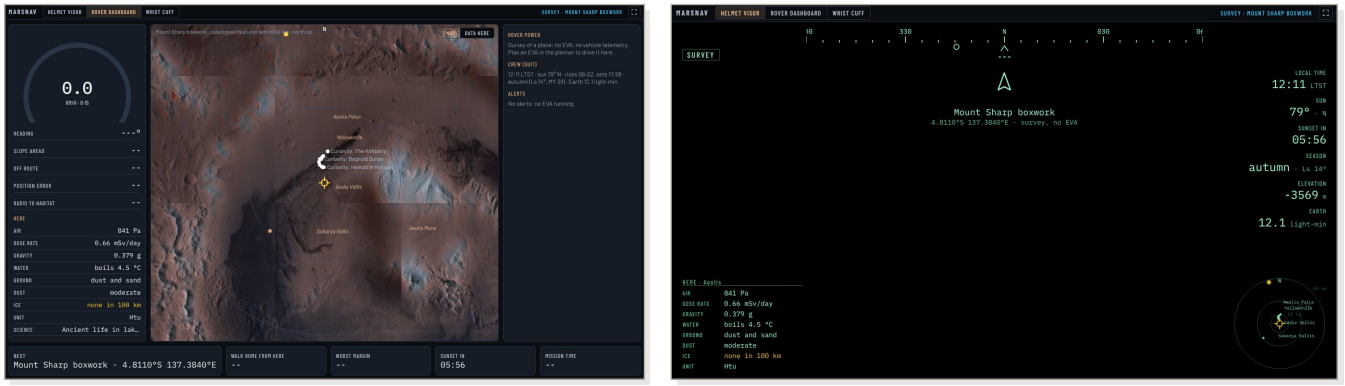
Lower: procedure, commands, consumables, cue card

Figure 40. *The wrist cuff during the EVA: big numbers for O_2, walk home and margin; the next stop with the rover's photographs of it; the map; the ground here as tiles with the nearest catalogued features; the station checklist; the commands; the consumables; the cue card; and every layer, folded.*

- Large targets for gloves. **ACK ALERTS** and **SILENCE** act on the planner's caution and warning, **DROP PIN** marks the crew's position with its error, and **RETURN NOW** is two steps: press, then confirm within five seconds.
- The header gives mission time, the mode and the local time on the mission clock.

A survey of any place

Opened from Landing mode or from the planner's place card, the displays show a place with no EVA running. The visor gives the local solar time, the sun, the time to sunset, the season, the elevation and the Earth's radio delay; the dashboard and the visor map the catalogued features and names within 50 km on Mars imagery, with range rings and the sun's direction; the cuff lists the facts, the nearest features and every layer.



Rover dashboard Helmet visor

Figure 41. *A survey of the Mount Sharp boxwork, opened from Landing mode: no EVA, the features within 50 km on Mars imagery, and the facts about the place.*

During an EVA the planner sends every layer at the crew's position again after 400 m of travel or ten minutes, and the site's terrain under the crew once a second. Without the MarsNav engine (the planner served as plain web files) the displays still follow the EVA, without the place facts.

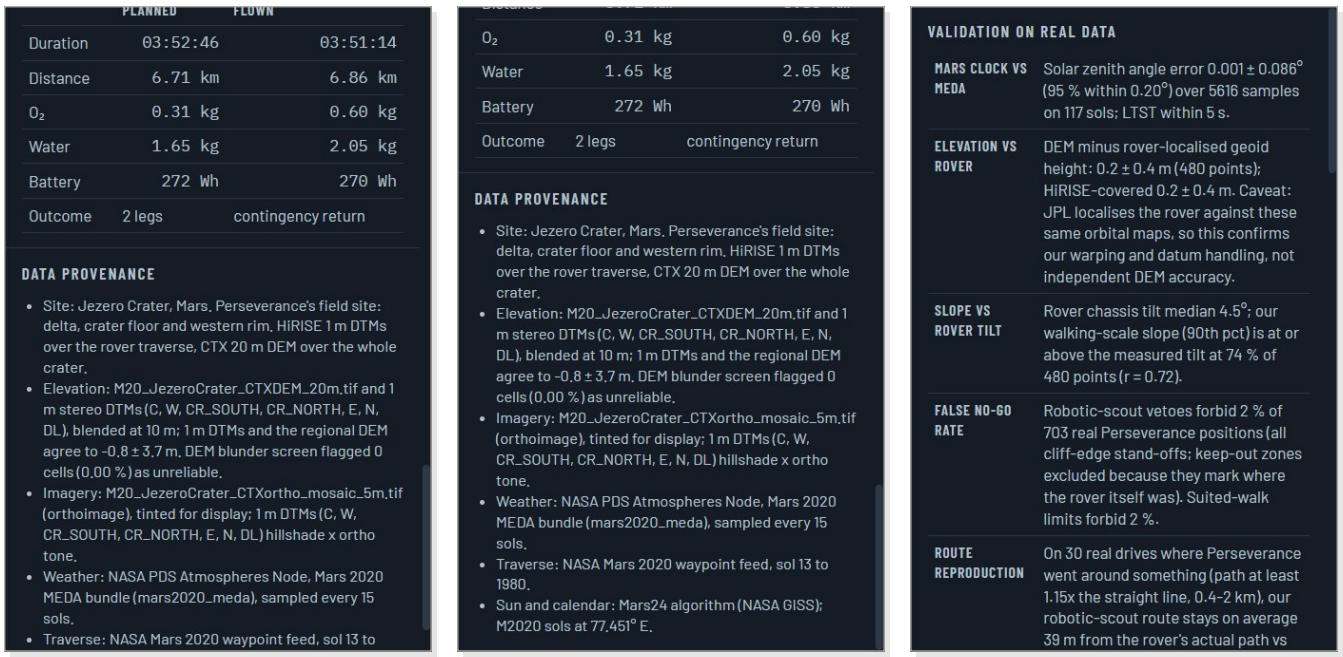
CHAPTER 11 · AFTER THE EVA

EVA planner: Review

Review is the debrief: what was planned against what was flown, every event, the paper fallback, files for other tools, and the software's own scorecard against real rover data.



Figure 42. *Review after an EVA cut short by a suit leak: exports and the event log (left), validation on real data (right).*



	PLANNED	FLOWN
Duration	03:52:46	03:51:14
Distance	6.71 km	6.86 km
O_2	0.31 kg	0.60 kg
Water	1.65 kg	2.05 kg
Battery	272 Wh	270 Wh
Outcome	2 legs	contingency return

DATA PROVENANCE

- Site: Jezero Crater, Mars. Perseverance's field site: delta, crater floor and western rim. HiRISE 1 m DTMs over the rover traverse, CTX 20 m DEM over the whole crater.
- Elevation: M20_JezeroCrater_CTXDEM_20m.tif and 1 m stereo DTMs (C, W, CR_SOUTH, CR_NORTH, E, N, DL), blended at 10 m; 1 m DTMs and the regional DEM agree to -0.8 ± 3.7 m. DEM blunder screen flagged 0 cells (0.00 %) as unreliable.
- Imagery: M20_JezeroCrater_CTXortho_mosaic_5m.tif (orthoimage), tinted for display; 1 m DTMs (C, W, CR_SOUTH, CR_NORTH, E, N, DL) hillshade x ortho tone.
- Weather: NASA PDS Atmospheres Node, Mars 2020 MEDA bundle (mars2020_meda), sampled every 15 sols.
- Traverse: NASA Mars 2020 waypoint feed, sol 13 to

O_2	0.31 kg	0.60 kg
Water	1.65 kg	2.05 kg
Battery	272 Wh	270 Wh
Outcome	2 legs	contingency return

DATA PROVENANCE

- Site: Jezero Crater, Mars. Perseverance's field site: delta, crater floor and western rim. HiRISE 1 m DTMs over the rover traverse, CTX 20 m DEM over the whole crater.
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- Imagery: M20_JezeroCrater_CTXortho_mosaic_5m.tif (orthoimage), tinted for display; 1 m DTMs (C, W, CR_SOUTH, CR_NORTH, E, N, DL) hillshade x ortho tone.
- Weather: NASA PDS Atmospheres Node, Mars 2020 MEDA bundle (mars2020_meda), sampled every 15 sols.
- Traverse: NASA Mars 2020 waypoint feed, sol 13 to 1980.
- Sun and calendar: Mars24 algorithm (NASA GISS); M2020 sols at 77.451° E.

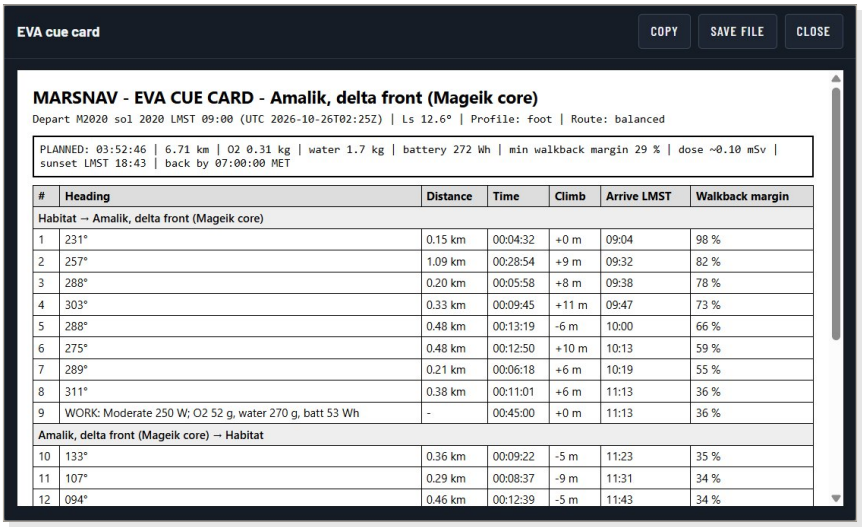
VALIDATION ON REAL DATA

MARS CLOCK VS MEDA	Solar zenith angle error 0.001 ± 0.086° (95 % within 0.20°) over 5616 samples on 117 sols; LTST within 5 s.
ELEVATION VS ROVER	DEM minus rover-localised geoid height: 0.2 ± 0.4 m (480 points); HiRISE-covered 0.2 ± 0.4 m. Caveat: JPL localises the rover against these same orbital maps, so this confirms our warping and datum handling, not independent DEM accuracy.
SLOPE VS ROVER TILT	Rover chassis tilt median 4.5°; our walking-scale slope (90th pct) is at or above the measured tilt at 74 % of 480 points (r = 0.72).
FALSE NO-GO RATE	Robotic-scout vetoes forbid 2 % of 703 real Perseverance positions (all cliff-edge stand-offs; keep-out zones excluded because they mark where the rover itself was). Suited-walk limits forbid 2 %.
ROUTE REPRODUCTION	On 30 real drives where Perseverance went around something (path at least 1.15x the straight line, 0.4–2 km), our robotic-scout route stays on average 39 m from the rover's actual path vs

Planned vs flown | Data provenance | Validation on real data

Figure 43. *The after-action comparison (duration, distance, O_2, water, battery, outcome), the provenance of every layer, and the validation scorecard against Perseverance.*

- **After-action**: planned versus flown for time, distance and every consumable, and the outcome (here a contingency return), linked to the re-plans in the event log.
- **Event log**: every alert, re-plan, pin and command with its mission time.
- **EVA cue card**: the paper fallback in the Artemis style, with heading, distance, time, climb, arrival time and walk-home margin for every leg. **Save file** saves it for printing; **Copy** copies it.
- **Routes as GeoJSON**: for MMGIS or QGIS.
- **Validation on real data** and **data provenance**: which NASA and USGS products, and which processing, produced each layer.



EVA cue card — COPY | SAVE FILE | CLOSE

MARSNAV - EVA CUE CARD - Amalik, delta front (Mageik core)

Depart M2020 sol 2020 LMST 09:00 (UTC 2026-10-26T02:25Z) | Ls 12.6° | Profile: foot | Route: balanced

PLANNED: 03:52:46 | 6.71 km | O2 0.31 kg | water 1.7 kg | battery 272 Wh | min walkback margin 29 % | dose ~0.10 mSv | sunset LMST 18:43 | back by 07:00:00 MET

#	Heading	Distance	Time	Climb	Arrive LMST	Walkback margin
Habitat → Amalik, delta front (Mageik core)						
1	231°	0.15 km	00:04:32	+0 m	09:04	98 %
2	257°	1.09 km	00:28:54	+9 m	09:32	82 %
3	288°	0.20 km	00:05:58	+8 m	09:38	78 %
4	303°	0.33 km	00:09:45	+11 m	09:47	73 %
5	288°	0.48 km	00:13:19	-6 m	10:00	66 %
6	275°	0.48 km	00:12:50	+10 m	10:13	59 %
7	289°	0.21 km	00:06:18	+6 m	10:19	55 %
8	311°	0.38 km	00:11:01	+6 m	11:13	36 %
9	WORK: Moderate 250 W; O2 52 g, water 270 g, batt 53 Wh	-	00:45:00	+0 m	11:13	36 %
Amalik, delta front (Mageik core) → Habitat						
10	133°	0.36 km	00:09:22	-5 m	11:23	35 %
11	107°	0.29 km	00:08:37	-9 m	11:31	34 %
12	094°	0.46 km	00:12:39	-5 m	11:43	34 %

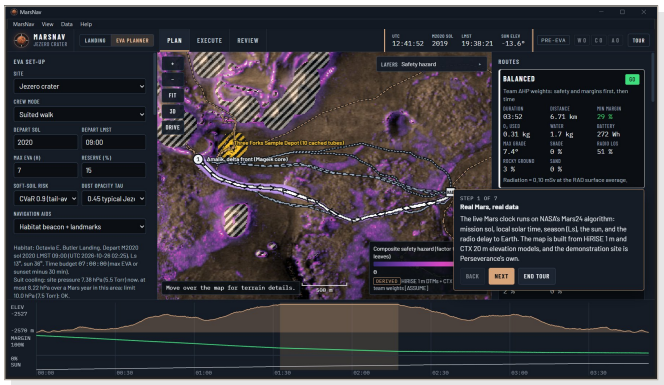
Figure 44. *The EVA cue card: the plan on paper if every display fails.*

CHAPTER 12 · SEVEN STEPS FOR FIRST-TIME USERS AND JUDGES

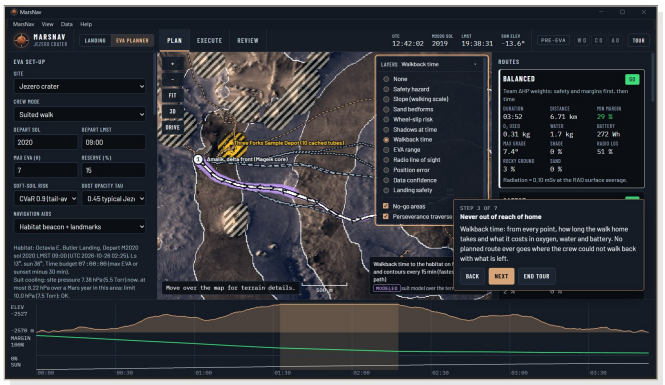
The guided Tour

Press **Tour** in the top bar at any time. The Tour drives the app itself through a real scenario and explains each step in a card; **Next**, **Back** and **End tour** control it.

Step	Title	What it shows
1	Real Mars, real data	The live Mars clock (Mars24), and a map built from HiRISE 1 m and CTX 20 m elevation models.
2	The ground a boot meets	Walking-scale slope and hazards from 1 m stereo terrain.
3	Never out of reach of home	Walk-home time from every point; no planned route ever goes where the crew could not walk back.
4	Five named routes	Balanced, safest, fastest, lowest consumption and fallback, with their budgets.
5	Walk it before you walk it	The 3D eye-height rehearsal on the real terrain.
6	Flying the EVA	Telemetry in NASA's SUITS format, a crew working harder than planned, a suit leak and the automatic walk-home.
7	Checked against Perseverance	The validation numbers against the rover's own records.

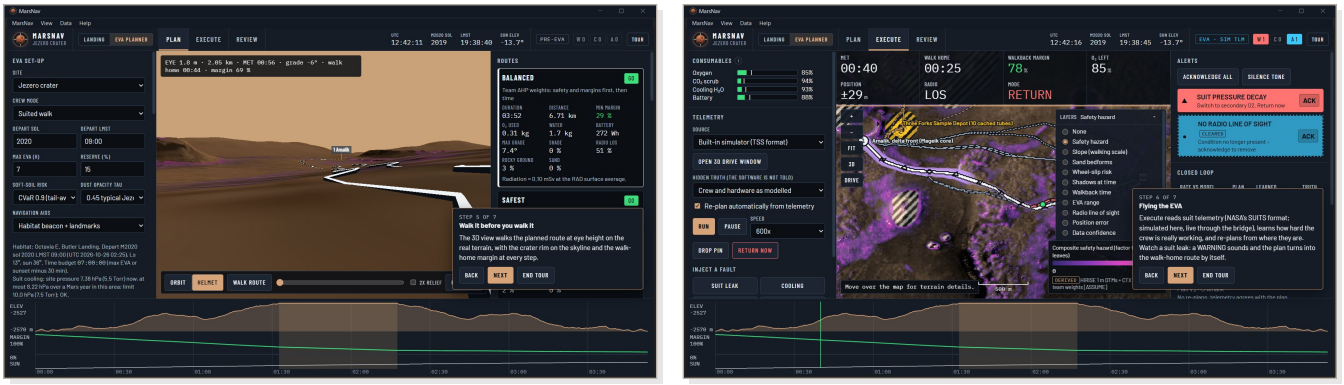


1 · Real Mars, real data



3 · Never out of reach of home

Figure 45. *Tour steps 1 and 3.*



5 · Walk it before you walk it

6 · Flying the EVA

Figure 46. *Tour steps 5 and 6: the 3D rehearsal, and Execute with a suit-leak warning and the return.*

CHAPTER 13 · THE SAME PLANNER ANYWHERE ON MARS

Other sites and new hubs

Everything specific to a place lives in one site file: area, projection, habitat, source data, mission calendar, targets and keep-outs. So the same planner runs on any site, including hubs you build in Landing mode.

jezero_east: a site with 20 m data only

Most of Mars will never have 1 m stereo terrain. **jezero_east** is built from a 20 m CTX elevation model alone to show the planner working at that resolution. Open it from **MarsNav** menu, **Jezero east demo site**. Every 1 m statistic is left empty, the hazard model falls back to calibrated 20 m estimates, and the planner says so: *regional DEM only: lower confidence*.

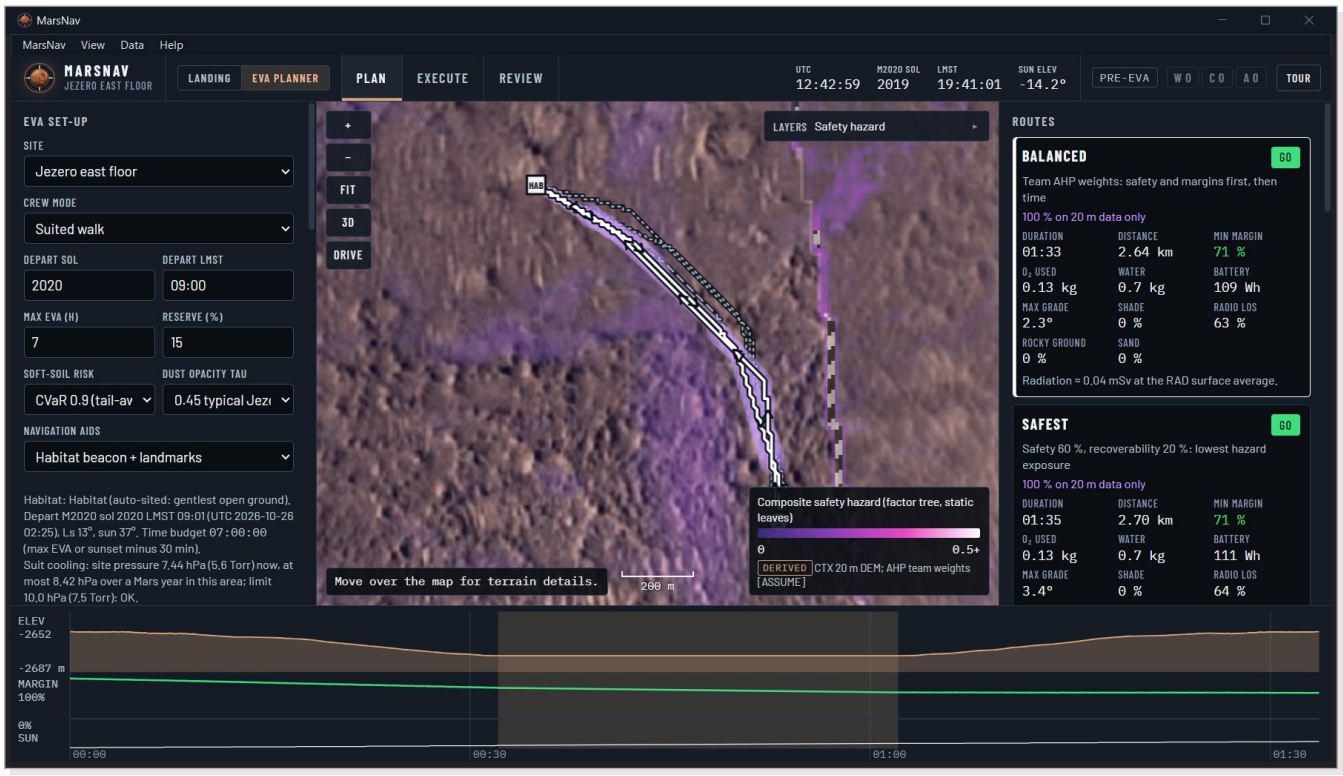


Figure 47. *jezero_east: five routes planned in 1.7 s on 20 m terrain, with the lower data confidence stated.*

Mount Sharp boxwork: the challenge's own example

The challenge names one place: the boxwork formations Curiosity studied on Mount Sharp, in Gale crater, in 2025. MarsNav has it as a full EVA site, built from five HiRISE 1 m elevation models (94 % of the operations area) and CTX 20 m models. Curiosity's track and its boxwork drill sites, Altadena, Monte Grande and Campo Marte, are placed at the rover's own positions, the mission clock is Curiosity's, and each drill site has the rover's photographs.



Plan 3D rehearsal

Figure 48. *The Mount Sharp site: the forward camp (HAB) and the walk to Altadena on the safety-hazard map, with cliffs and buttes hatched (left); the same walk in 3D on 1 m terrain (right).*

MarsNav's answer here is a finding. Cliff-bounded buttes (every one at least 8 m high) and 39 % sand leave much of the foothills out of reach on foot with dead-reckoning navigation, so the hub is a forward camp in a pressurised rover at Curiosity's sol 4530 stop. From there a suited crew reaches Altadena in 1.3 h and Campo Marte in 3.4 h.

Hubs built in Landing mode

A hub committed in Landing mode (Chapter 4) becomes a site like any other: choose it in the planner's **Site** selector or press **Open** in the Hubs tab. It has its own terrain, hazard layers, imagery, weather, targets, 3D view, drive window and closed loop.

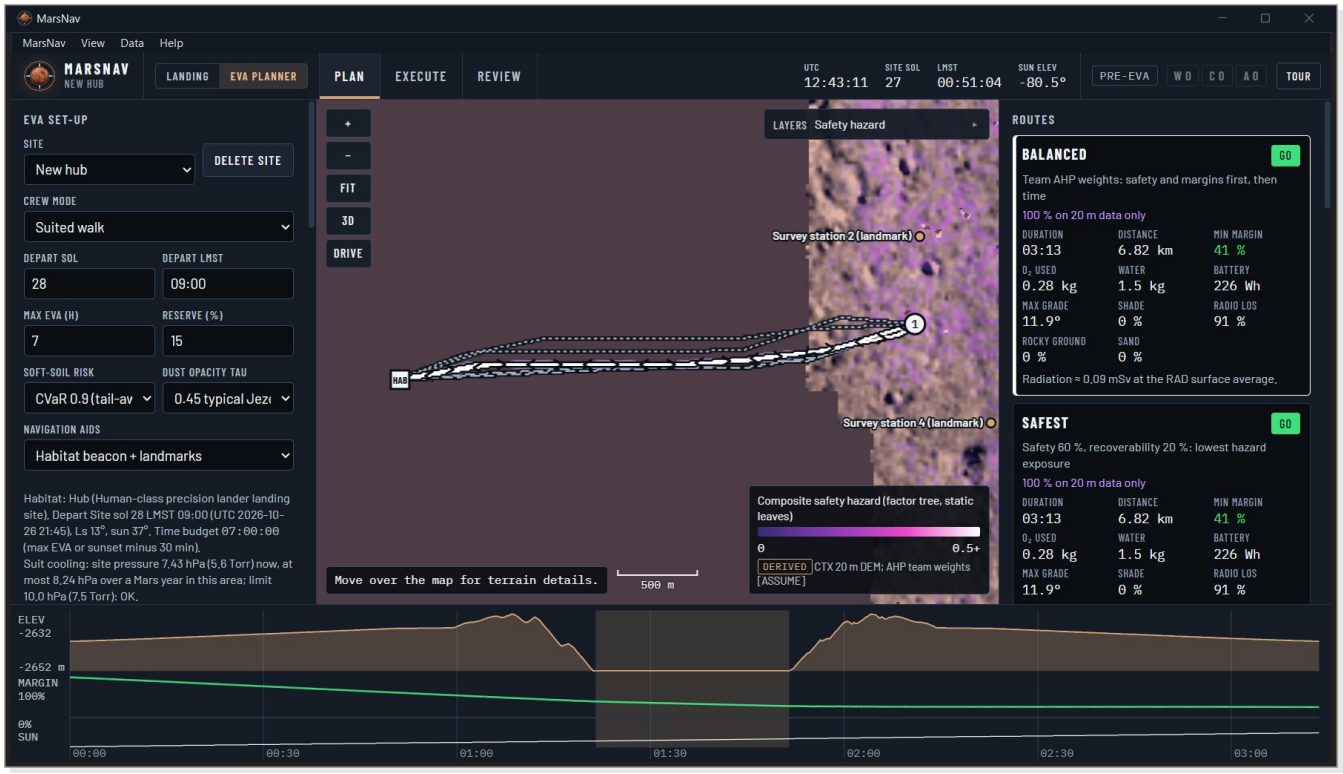


Figure 49. *A hub built in Landing mode near the equator (3.1°N, 155.3°E): CTX 20 m elevation blended into MOLA, survey stations around the habitat, and routes planned in 4.0 s.*

CHAPTER 14 · CLOSING THE LOOP WITH A REAL SUIT

Live telemetry and hardware

During the EVA, MarsNav reads suit and rover telemetry in the format of NASA's SUITS Telemetry Stream Server (TSS), the server NASA uses for its spacesuit-interface challenge, learns from it and uplinks every re-plan. The telemetry bridge is built into the app: there is nothing to install or start.

In **Execute**, set **Telemetry** to **Live suit telemetry** and press **Enable live telemetry**:

- **Automatic** (default): MarsNav looks for a TSS at the last address used, then on this computer (port 14141), then with a UDP broadcast on every local network. It detects the TSS year (2026 or 2027), polls it once a second and streams the frames to the app. If none answers within about 1.6 s it switches to the built-in **demo suit** and says so.
- **A TSS at a specific address**: type its IP (or IP:port); it is remembered for next time.
- **Demo suit**: a simulated crew member flies the plan with the hidden truth you choose (working harder, walking slower, battery drain, a leak) and follows every re-plan, as a real crew would.
- Every re-plan is uplinked at `/api/tss/plan` for a helmet display; `/api/tss/status` shows the link. Only TSS read commands are ever sent.

> **Tested against NASA's own server**
> MarsNav was driven over UDP by a TSS2027 build compiled from NASA's source. The closed loop learned that the test-week suit drains its battery and CO_2 bed more than eight times faster than the model, and turned the EVA home within about two minutes.

Navigation without GPS

Position comes from an extended Kalman filter built on Mars practice: dead reckoning from steps or wheel odometry, sun heading, map-matching against landmarks extracted from the terrain, and ranging to the habitat beacon. Its 2σ error is drawn on the map as a radius (never a false exact dot) and feeds the planner: the stand-off from cliffs and keep-outs grows where the crew will be less sure where they are.

CHAPTER 15 · THE SCIENCE AND ENGINEERING INSIDE

How MarsNav works

MarsNav is three things working together: a data pipeline that turns orbital measurements into the ground a boot meets, a planning engine that puts safety first, and a whole-planet engine that streams and screens Mars. Every constant in the code is tagged by its origin: [SRC] sourced, [CALIB] calibrated, [PHYS] physics, [ASSUME] assumption.

Technology

Layer	Built with
Desktop app	Python launcher compiled with PyInstaller (tools/build_windows.py, the same recipe locally and in CI); native window through pywebview on Microsoft Edge WebView2 (pythonnet / .NET); a supervisor with a Windows job object, health checks and automatic restarts.
Interface	HTML, CSS and JavaScript modules, no framework; three.js for 3D; the planning engine in a Web Worker; the crew displays (hud.html) fed over a BroadcastChannel; one renderer for the data at any place (placeview.js).
Planning engine	Plain JavaScript (engine/): terrain, cost model, planner, walk-home fields, physiology, station tasks, Mars clock, environment, AHP, risk, sortie selection (Held-Karp, QUBO), simulation, closed loop, alerts, procedures, reports. The same code runs in the app, the tests and the validation.
Planet engine and server	Python (planet/): pooled HTTP server, tile pyramid and disk cache, MOLA / HRSC / CTX / HiRISE readers over GDAL in worker processes, USGS STAC catalogue, landing assessment, site generator, Local mode, TSS bridge, the science atlas, missions and objectives, space weather, and the lookup of every layer at a point.
Data pipeline	Python with numpy, scipy, rasterio / GDAL and Pillow (pipeline/): download with SHA-256 checks, terrain statistics, hazard layers, imagery, weather fit.
Verification	Node's test runner (64 engine and closed-loop tests), pytest (53 pipeline, downloader, science-atlas and place tests), validation scripts against Perseverance data, an end-to-end server self-test, and on every change a Windows build that is started and checked before it is published (GitHub Actions).

From orbit to the ground a boot meets

- **Grids.** Each site gets a 10 m operations grid, a 40 m horizon grid and a 100 m regional grid in a site-centred projection (scale error under 0.1 %).
- **Data screens.** Stereo blunders (spikes, pits, striping) are flagged and vetoed as unreliable; mosaic seams are found and repaired. The screen found and fixed a real 18 m seam that had cut one site in two with a false cliff.
- **Walking-scale statistics from 1 m HiRISE models**, per 10 m cell: walking slope (90th percentile on a ~4 m baseline, one or two strides), step height (99th percentile of detrended relief), roughness, and a bedform detector using orientation coherence: ripples and TARs are directional and tall, rubble is neither.

- **20 m-only ground** gets walking slope from a conservative calibration fitted where both resolutions overlap, moderate priors for steps and roughness, and zero data confidence, so the planner says so.
- **Other layers**: cliff distance, landmarks for navigation, horizons in 16 directions out to 25 km with Mars curvature, keep-outs (sample depot, active rover, planetary protection), and science value.

Safety first, then optimisation

Hard limit (veto)	Rule
No reliable terrain data	Outside the elevation model, or a flagged blunder.
Keep-out, planetary protection	Within 2σ of position error of a keep-out or Special Region.
Cliff	Cliff distance minus 2σ below the stand-off (10 m on foot, 20 m by rover).
Slope	Walking slope over 20° on foot; 25° on rock or 12° on sand by rover; 30° for a robot.
Step	Over 0.8 m on foot; 0.35 m by rover.
Soft soil	Wheel-slip CVaR (the mean of the worst 10 % of outcomes) at or above 0.7 for vehicles.
Walk home	Any partial route from which the crew could not walk home with what is left is dropped during the search.

Only then are routes optimised, over five criteria each normalised by the fastest route: time, hazard exposure, consumables, recoverability (walk-home time, radio line of sight, position error) and science value missed. The default weights come from the team's pairwise judgements with AHP (safety 0.42, consumables 0.22, recoverability 0.22, time 0.09, science 0.04; consistency ratio under 0.01). NASA has published no Mars route weights, so they are documented as an assumption and shown to be robust: 40 random perturbations keep 100 % of runs within 80 % of the same route.

Crew, suit and the walk-home field

- **Walking speed** from Apollo planning speeds, slowed on slopes by an Apollo regolith footing curve and a constant-metabolic-power limit; slower on sand and rocky ground.
- **Metabolic power** from the Minetti cost of transport scaled to Mars gravity, with suit inefficiency; oxygen, CO_2 and heat follow by physics, plus a small suit leak.
- **Walk-home fields.** For every point, the time and consumables needed to walk home are computed over the whole map, so the search can test the guarantee instantly at every step.
- **Environment**: the Mars24 clock, sun and shadows from horizon maps, temperatures from a harmonic fit to MEDA records, dust opacity, and radiation from the Curiosity RAD surface average.

The whole planet

Zoom	Elevation	Imagery
Whole planet	MOLA MEGDR 16 px/deg (3.7 km)	Viking MDIM 2.1 colour, 232 m

Zoom	Elevation	Imagery
Regional	MOLA MEGDR 128 px/deg (463 m); HRSC-MOLA blend 200 m	Viking colour, THEMIS 100 m
Close	CTX stereo models, 20 m (44,628 on USGS STAC)	CTX global mosaic, 5 m
Closest	HiRISE stereo models, 1 m (1,262 on USGS STAC)	HiRISE, 0.25-1 m

Mars is cut into a pyramid of map tiles, the same tiling as NASA's Mars Trek. Each frame the globe keeps only the tiles in view; a tile splits into four finer ones when it covers more than about 330 pixels, and a coarse tile keeps drawing until its children arrive, so the ground never has holes. The local server builds each tile on demand from the best data under it and keeps a disk cache (20 GB by default, oldest removed first).

On top of the elevation and imagery sit the science atlas (bundled catalogues of 74,250 mapped features, grids installed once, NASA Mars Trek colour maps streamed, and models of pressure, dose and sunlight from MOLA), and one lookup that gathers every layer at a point (`planet/place.py`, `/api/place`) for Landing mode, the planner's place card and the crew displays (Chapter 5).

CHAPTER 16 · CHECKED AGAINST PERSEVERANCE

Validation and testing

MarsNav is tested against the real rover's records, not only against itself, and every result is reported, including the one that did not go our way. The figures below were re-run on 26 October 2026 for this guide.

Check	Result
Mars clock vs Perseverance MEDA sun geometry	Solar zenith angle error 0.001 ± 0.086° (95 % within 0.20°) over 5,616 samples on 117 sols; local true solar time within 5 s.
Terrain vs the rover	Walking-scale slope at or above the rover's measured chassis tilt at 74 % of 480 points (r = 0.72). Elevation agrees to 0.2 ± 0.4 m, but that check is not independent: JPL localises the rover against the same orbital maps.
False no-go rate	2 % of 703 real Perseverance positions, all cliff-edge stand-offs.
Hazard model vs JPL's choices	On 30 detour drives the hazard index along the rover's real path is 0.182 against 0.208 for the straight line; the real path scores lower on 21 of 30. MarsNav's risk-aware route scores 0.111.
Bedform avoidance	Strong bedform evidence on 3 % of the rover's path against 7 % of the surrounding 1 km corridor.
Route geometry vs the rover's drives	Not better than a straight line (38 m vs 34 m mean offset). Reported as a negative result: rover drives are shaped by science stops, AutoNav and sub-metre hazards.
Same trip as the rover	Séítah to the delta front: Perseverance drove 7.1 km over 85 sols; MarsNav plans 3.07 km for a robotic scout in under a second, or 2.92 km in a 1.4 h EVA for a crew. Over 45 real drives its routes cross 47 % less mapped sand and 55 % less rocky ground than the rover did, and its speed model reproduces the rover's measured 79.7 m/h (79 m/h). It rates Séítah only 1.5x slower per metre where JPL projected 4x, so for a rover it still prefers the shorter corridor JPL rejected.
Weight robustness	AHP consistency ratios all under 0.01; 40 random perturbations keep 100 % of runs within 80 % of the balanced route.
Closed loop vs flying the filed plan	40 paired EVAs with crews and hardware the planner was not told about: 0 of 40 dipped into the reserve with the loop, 31 of 40 without (Wilcoxon $p < 0.001$).
Against NASA's TSS2027 server	Learned an 8x faster battery and CO_2 drain and turned the EVA home within about 2 minutes.
Station selection	Held-Karp exact over all 32,767 sets of 15 candidates in about 35 ms.
Speed	2.3 million 10 m cells; five named routes in about 3 s on one laptop core.
Reproducibility	A full rebuild from the raw USGS files gives byte-identical layers.

The closed-loop campaign

Measure	Closed loop	Open loop (filed plan)
Lowest usable consumable left, mean	24.6 %	−5.1 %
Lowest usable consumable left, worst run	3.1 %	−40.8 %
Runs that dipped into the reserve	0 of 40	31 of 40
Runs that needed a RETURN NOW warning	0	31
Science completed without touching the reserve	0.64	0.29

Each EVA is flown twice with the same random numbers and a hidden truth drawn per run (work rate ×0.9-1.4, cooling water ×0.9-1.4, battery drain ×0.9-1.5, walking pace ×0.7-1.1). The open-loop crew completes more raw science (0.90 vs 0.64) because it often finishes stations on reserve oxygen or battery; counting only science done without touching the reserve, the loop comes out ahead. Both arms use the same simulator.

Automated tests

Suite	Command	Result
Engine and closed loop (Node 20+)	node --test tests/*.test.mjs	64 of 64 pass (29 Oct 2026)
Pipeline, downloader, science atlas, place	python -m pytest tests -q	53 of 53 pass (29 Oct 2026)
Validation vs Perseverance	node validation/validate.mjs	results above (26 Oct 2026), about 30 s; needs the raw NASA inputs in data_cache
Closed-loop campaign	node validation/closed_loop_campaign.mjs	0 / 40 vs 31 / 40; reproduced exactly on 29 Oct 2026
Same trips as the rover	node validation/rover_case_study.mjs	results above (26 Oct 2026), about 15 s
Desktop server under load	selftest.bat (MarsNav.exe --selftest)	every check passes on a clean Windows machine: about 10,000 requests, a worker and the server killed, the server replaced in 3 s, no process left
The built program	python tools/smoke_exe.py dist\MarsNav\MarsNav.exe	16 pages and scripts, 9 APIs and the place and elevation lookups answer
On every change	.github/workflows/windows-exe.yml	builds MarsNav.exe on a clean Windows machine, runs the two checks above and the test suites, then publishes the zip
Browser tour (Playwright)	every site, Landing mode, the drive window and the crew displays	29 of 29 checks, no console errors (29 Oct 2026)

The tests include the planner's safety contract on real terrain (every route inside its hard limits with a walk-home plan at every point), navigation-filter consistency, estimator convergence, the UDP telemetry bridge over real sockets, exact against heuristic station selection, the science catalogues against brute force, the dose model against Curiosity's value at Gale, the place lookup with no data and no network, the USGS chart at every landed mission, gravity and the liquid-water window against published values, and the Mars Year at its published starts.

Full reports: `validation/REPORT.md`, `validation/CLOSED_LOOP.md`, and the method in `docs/05_verification_and_validation.md`.

CHAPTER 17 · FOR JUDGES

How MarsNav answers the challenge

The challenge asks for an interplanetary survival guide: a layered map that helps astronauts plan and carry out a safe, successful Marswalk with science along the way. Here is MarsNav against the challenge and the criteria.

What is new compared with existing tools

Mapping tools show terrain; route planners optimise a cost. MarsNav makes NASA's walkback rule an algorithm, prices the uncertainty that trapped Spirit, treats navigation error as a planning input, closes the loop with suit telemetry, and carries the same safety logic from choosing the landing site to the last step of each EVA, offline.

The challenge asks for	MarsNav
"a layered, integrated view of a location or route"	Eleven terrain layers and four overlays at 10 m, routes with uncertainty, a timeline profile; the whole planet with six base maps, six overlays and a 26-layer science atlas; every layer at any point, one click (Chapter 5).
"at least three kinds of data": optical, thermal, spectral, topographic, geophysical, atmospheric, radiation, chemical	All eight: HiRISE, CTX and Viking images; TES and THEMIS thermal; CRISM, OMEGA and TES minerals; MOLA, HRSC, CTX and HiRISE elevation; MRO gravity; air pressure; cosmic-ray dose; GRS elements (Chapter 6).
"data from multiple NASA science missions", past and future	MRO, MGS, Mars Odyssey, Mars Express, Mars 2020, MSL, Viking and Apollo data in the map, and a catalogue of 31 missions past, present and future.
science goals and a location: the boxwork at Mount Sharp	Seven science objectives that rank every site, and the Mount Sharp boxwork as a full EVA site with Curiosity's drill sites (Chapters 6, 13).
space weather	NASA DONKI CME arrivals at Mars, particle events, flares and an EVA advisory; dose by altitude on every route.
"details of their routes and destinations"	Guidance legs, station procedures, target provenance, the rover's own photographs of each stop, the cue card, GeoJSON.
"updates on current conditions"	Live Mars clock, sun and shade along the route, dust, radio line of sight, position error, space weather, and live suit telemetry with learned rates.
"complete their mission quickly and safely"	Five named routes under hard limits, walk-home margins everywhere, caution and warning, automatic return.
"plan and carry out a successful Marswalk"	Plan, Execute and Review, with the 3D rehearsal and the drive window.
inside the habitat, on a tablet, or a HUD on the helmet visor	The habitat console, a helmet-visor HUD, a rover dashboard and a wrist cuff, live from the same EVA (Chapter 10).
point-of-view images	The rover's own Navcam and Mastcam frames from each station (Chapter 8).
"new and exciting science along the way"	Perseverance's real sample sites, a science-value optimiser, what is catalogued within 100 km of every site, landing sites scored for water ice.

The judging criteria

Criterion	Evidence in MarsNav
Influence	The decisions NASA must make before the first crewed EVA; site-agnostic; reads NASA's SUITS telemetry format; runs offline on one laptop; open data and open source.
Creativity	Walk-home guarantee inside the search; soft soil priced by its tail (CVaR); position error as a planning input; walking-scale terrain from 1 m stereo; a closed loop that learns the crew and the suit; every layer at any point; the same EVA on visor, dashboard and cuff; Mars mode.
Validity	Validated against Perseverance; a negative result reported; 117 automated tests; a paired 40-EVA campaign with a significance test; byte-identical rebuilds; a Windows build tested on a clean machine at every change; assumptions tagged.
Relevance	Every point of the challenge statement (above), demonstrated at Perseverance's own field site.
Presentation	A native desktop app with a guided Tour, 3D rehearsal and three crew displays; this guide; focused documents in docs/.

CHAPTER 18 · FOR DEVELOPERS AND REVIEWERS

Building from source

The portable app is built from this repository with one script. You only need this chapter to change the code, rebuild the science data or reproduce the results.

Build the desktop app

On Windows with Python 3.10 or later (tick "Add python.exe to PATH") and an internet connection, double-click **build_exe.bat** in the project folder. It:

1. creates a private build environment in `.buildvenv` (your own Python is not touched);
2. installs numpy, scipy, rasterio, Pillow, pywebview, pythonnet and PyInstaller from `requirements.txt`;
3. compiles `launcher/martian_map.py` and `launcher/desktop.py` into **MarsNav.exe**, a windowed program with its libraries in `MarsNav_runtime`, with `tools/build_windows.py` (the recipe the published build uses);
4. assembles the portable package: `dist\MarsNav\` and `dist\MarsNav-Windows.zip`;
5. starts the new MarsNav.exe as a check.

The `app/` and `planet/` folders are read from disk, so changes there need no rebuild; changes to `launcher/` or the pipeline do.

The published download

Every change pushed to the repository runs `.github/workflows/windows-exe.yml` on GitHub Actions. On a clean Windows machine it builds MarsNav.exe with the same recipe, starts the built program and checks every page and API (`tools/smoke_exe.py`), and runs the program's own self-test under load. On Linux it runs both test suites and the closed-loop campaign. Only when all of them pass does it publish **MarsNav-Windows.zip** as the release of the current version, replacing the previous build of that version.

Run from source

```
# the desktop app from source (Windows, with the build environment)
.buildvenv\Scripts\python.exe launcher\martian_map.py

# text modes
MarsNav.exe --console      # the text menu, app in the default browser
MarsNav.exe --headless     # serve only (development, tests)
MarsNav.exe --selftest     # end-to-end server check under load
```

Rebuild the science data

The data ship pre-built. To regenerate them from the raw NASA and USGS files (about 0.9 GB download, about 10 minutes), use **Data** menu, **Rebuild Jezero data**, or from source:

```
pip install -r requirements.txt
bash tools/build_all.sh     # fetch + SHA-256 check, terrain, layers, weather fit, imagery
```

Raw rasters are read from data_cache/raw/, a _data_raw/ folder next to the project, or $MM_RAW. Exact URLs and checksums are in pipeline/sources.json. A full rebuild reproduces the shipped layers byte for byte.

Tests: node --test tests/*.test.mjs (64), python -m pytest tests -q (53), node validation/validate.mjs, node validation/closed_loop_campaign.mjs, python tools/smoke_exe.py, and selftest.bat (Chapter 16).

CHAPTER 19 · IF SOMETHING DOES NOT WORK

Troubleshooting

Most problems have a one-line fix. MarsNav also keeps logs that say exactly what happened.

Symptom	What to do
"Windows protected your PC"	Expected for an unsigned program. Click More info, then Run anyway. Antivirus may also ask once.
"MarsNav cannot find its 'app' folder"	The exe was moved on its own. Keep MarsNav.exe with the folders it came with; move the whole folder.
"MarsNav cannot save data in its folder"	The folder is read-only (for example inside C:\Program Files). Copy the whole folder to your Desktop, Documents or another drive.
The window opens in an Edge window instead	This PC has no WebView2 runtime. Everything works the same; install WebView2 from Microsoft to get the native window.
Starting it again does nothing	MarsNav is already running: its window is brought to the front. Look for it in the taskbar.
The first start is slow	Windows checks the program files the first time; later starts take a few seconds.
Landing mode is blurry close up	Close-up detail needs CTX (20 m) or HiRISE (1 m) data under that spot; elsewhere 463 m MOLA is the best that exists. The first view of an area downloads its data in the background and sharpens by itself.
A science layer did not install	Each comes from a different server. The others work; press Install the missing layer later in the Data tab.
Local mode says it will not fit	Untick the 11 GB blended elevation or a THEMIS pack, use a smaller zone radius, or free disk space.
The map stops responding	Wait a few seconds: the supervisor replaces a stalled engine within 30-60 s (3 s if it exits). Help, Engine status shows its health.
SPACE WX says there is no data	NASA's DONKI service could not be reached, or limited the demonstration key. Press Fetch the latest from NASA DONKI later, or set your own key in MM_NASA_API_KEY. The advisory never guesses.
A crew display says it is waiting	Open it from the EVA planner (HUD, or the buttons in Execute) in the same MarsNav, then start an EVA. In a browser, allow pop-ups for the page.
A line of the place data says reading...	A remote lookup (the USGS geologic map or a Mars Trek map) is still answering; it fills in by itself. In Mars mode it says the map is not in the local packs.

Logs and the self-test

Help, **Open the log folder** opens `data_cache`: `server.log` (start-up, restarts, errors and a memory census), `desktop.log`, `server_child.log`, `workers.log` and, if the engine ever stalled, `server_hang.log` with what every thread was doing. **selftest.bat** loads the server for about three minutes, kills a raster worker and the server on purpose, and checks that everything recovers; its report is `data_cache\selftest.txt`.

Keep some space free on C:

Windows needs free space on the system drive to grow its page file. With almost none, every program can run out of memory. MarsNav writes nothing to the system drive itself, but warns in its log when it is below 2 GB.

CHAPTER 20 · WHAT MARSNAV DOES NOT CLAIM

Honest limitations

A tool for life-critical decisions must be clear about its limits. These are condensed from docs/08_limitations.md; every assumption is tagged [ASSUME] in the code.

Claims not made

- **Not a flight product.** Not certified to any NASA standard. The display is informed by NASA-STD-3001 Appendix F and the SUITS requirements, not compliant with them.
- **No human has walked on Mars.** Walking speed (Apollo), metabolic cost (Minetti, scaled to Mars gravity), suit capacities (EMU heritage), the 20° slope limit and the 0.8 m step limit are assumptions carried over from Earth and Moon.
- **The route weights are the team's own.** NASA has published none; they are shown to be robust, not official.
- **Routes do not reproduce Perseverance's actual drives** better than a straight line. This is reported, not tuned away.
- **Landing-site scores are a first screen** from orbital data. A crewed site still needs HiRISE imaging and 1 m elevation over the whole ellipse and an engineering review. SWIM says where ice is consistent with every dataset, not that it has been sampled.

Data and software limits

- 1 m HiRISE coverage is 75 % of the Jezero operations area with all seven DTMs (64 % with the default four). Elsewhere the map is 20 m-based, and boulders or scarps smaller than about 20 m are not in it; route cards say so.
- The weather model is a fit to sampled MEDA records: typical conditions, not a forecast. Dust storms enter through .
- Radiation uses the Curiosity RAD surface average; solar particle events are a contingency, not a modelled dose.
- The closed loop has run against NASA's TSS2027 server, not flight hardware; the campaign measures the loop's logic with a shared simulator.
- The path search keeps up to four labels per grid cell (near-optimal across five criteria); returned routes are re-evaluated exactly afterwards.
- Voice is an offline intent parser with speech output (no speech recognition). The global traverse router plans long drives on 20-463 m elevation; build a hub at each end for EVA-scale planning.
- **Place values** describe the ground around a point at each dataset's resolution (GRS 5°, TES 3-16 km, SWIM 0.05°), not what a drill will meet. TES mineral and dust values are read back from published colour maps, so they are relative.

- **Space weather** is a stated heuristic on NASA's DONKI records, which see the Sun from Earth's side. The live service could not be reached from the development network on 28-29 October 2026, so the parser is tested on DONKI's documented record shapes. There is no crustal magnetic field layer: no openly licensed global grid was found.
- **The crew displays** take heading from the track, because SUITS telemetry has no compass, and simulate a HUD on a screen; they have not been tried on a visor.
- The Windows program is not code-signed, so Windows asks once before running it on a new PC.
- Moving the weather model to a new latitude is a first-order correction, not a climate model. MarsNav is a Windows program; the EVA planner alone runs in any modern browser from a simple web server (Chapter 18).

CHAPTER 21 · WHERE EVERYTHING IS

Project structure

The repository layout, for anyone reading or extending the code.

Path	Contents
`MarsNav.exe, MarsNav_runtime/, dist/`	The desktop app built by build_exe.bat, its libraries, and the portable folder and zip. Build output, kept out of the repository (.gitignore).
`launcher/`	martian_map.py (supervisor, local server, worker and job processes, self-test), desktop.py (the native window, menus, dialogs), wintheme.py (dark title bar and menu), make_portable.py, the icon and version resource.
`app/`	The interface: index.html (EVA planner), landing.html (Landing mode), drive.html (3D drive window), hud.html (crew displays), css/, js/ (placeview.js draws the data at any place), theme.css (the shared console style), engine/ (copy of engine/), data/ (built sites, station photographs), vendor/ (three.js and the fonts), sw.js (offline cache).
`engine/`	The planning engine in plain JavaScript, shared by the app, the tests and the validation.
`planet/`	The whole-planet engine: tiles and cache, raster readers, STAC catalogue, IAU names, landing assessment, site generator, science datasets, the science atlas (atlas.py, features.py), missions and objectives (science.py), space weather (spaceweather.py), every layer at a point (place.py), Local mode, TSS bridge and the HTTP API; data/ holds the bundled catalogues, candidates, missions and objectives.
`pipeline/`	The Python data pipeline and the site definitions (pipeline/sites/*.json).
`validation/`	Validation against Perseverance and the closed-loop campaign, with their reports.
`tests/`	Engine and closed-loop tests (Node); pipeline, downloader, science-atlas and place tests (pytest).
`tools/`	build_all.sh; build_windows.py and smoke_exe.py (the Windows build and its check); build_features.py (the bundled catalogues); build_pov.py (station photographs); the TSS bridge and stand-in; guide/ (the screenshot scripts and this guide's source); dev/ (developer scripts).
`.github/workflows/`	windows-exe.yml: builds, checks and publishes the Windows app on every change.
`docs/`	Concept of operations, architecture, algorithms, data provenance, verification, requirements traceability, deployment and new sites, limitations, the closed loop and hardware interface, station tasks, station selection.
`submission/`	Judging-criteria map, demo script, comparison with other approaches, screenshots.
`build_exe.bat, selftest.bat`	Build the app; test the server end to end.
`data_cache/`	Created at run time: downloads, tiles, science layers, Local mode packs, built sites, reports, logs. Not part of the repository.

CHAPTER 22 · STANDING ON NASA'S SHOULDERS

Data sources and credits

MarsNav is built entirely on public data. All mission data are US Government works in the public domain or openly licensed; exact sources, URLs and checksums are in pipeline/sources.json and docs/04_data_and_provenance.md.

Data	Source
HiRISE 1 m stereo DTMs and orthoimages; CTX 20 m DTMs and 6 m orthoimages	Mars Reconnaissance Orbiter; USGS Astrogeology (STAC catalogue), NASA-JPL Mars 2020 products
MOLA MEGDR elevation	Mars Global Surveyor (Smith et al. 2001), PDS Geosciences Node
HRSC-MOLA blended DEM, 200 m	Mars Express / MGS (Fergason et al. 2018), USGS
CTX global mosaic, 5 m	Dickson et al. 2023, Caltech Murray Lab; served by Esri / ASU
Viking MDIM 2.1, THEMIS infrared mosaics, Exploration Zone centres	USGS; ASU; NASA Mars Trek
MEDA weather, rover traverse and tilt	Mars 2020 Perseverance; PDS Atmospheres Node; Mars 2020 waypoint feed
RAD surface radiation	Mars Science Laboratory Curiosity
SWIM subsurface water-ice consistency	Morgan et al. 2021 (Nature Astronomy), 2025 (PSJ); Planetary Science Institute
TES thermal inertia	Putzig & Mellon 2007 (Icarus), PDS Geosciences Node
GRS water-equivalent hydrogen	Mars Odyssey (Boynton et al. 2007, JGR), PDS
Crater database (384,343 craters ≥ 1 km)	Robbins & Hynek 2012 (JGR), 2020 release
Global geologic map, SIM 3292	Tanaka et al. 2014, USGS (OGC API)
Candidate sites	NASA First Landing Site / Exploration Zone Workshop (2015); Golombek et al. 2021 (Starship); NSSDCA
Mineral catalogues: aqueous minerals, CRISM / OMEGA hydrous detections, chlorides	Ehlmann & Edwards 2014; Carter et al. 2013 (MRO CRISM, Mars Express OMEGA); Osterloo et al. 2010; via NASA Mars Trek
TES minerals, albedo and dust cover; MOLA roughness	Bandfield 2002; Christensen et al. 2001; Ruff & Christensen 2002; Kreslavsky & Head 2000; NASA Mars Trek
Glaciers, gullies, valley networks, fans, dunes, present-day activity	Souness 2012, Brough 2016, Fassett 2010; Harrison 2015; Hynek et al. 2010; Moore & Howard 2005, Kraal 2008; Hayward et al. 2007; Daubar, McEwen, Ojha, Dundas; via NASA Mars Trek
GRS elements (Fe, Cl, K, Th, Si); gravity field GMM-3	Mars Odyssey (Boynton et al. 2007), PDS; MRO (Genova et al. 2016), PDS Geosciences Node
Space weather	NASA CCMC DONKI (api.nasa.gov, kauai.ccmc.gsfc.nasa.gov)
Cosmic-ray dose and its pressure dependence	Curiosity RAD (Hassler et al. 2014; Guo et al. 2017)

Data	Source
Station photographs, rover positions	Perseverance and Curiosity raw images (NASA/JPL-Caltech, MSSS, ASU); NASA/JPL MMGIS waypoint feeds
Mission statuses	NASA, ESA, JAXA, CNSA and press reports, sourced per mission in planet/data/missions.json (28 October 2026)
USGS chart quadrangles	USGS 1:5,000,000 Mars Chart series (MC-01 to MC-30)
Place names	IAU Gazetteer of Planetary Nomenclature
Mars clock	Mars24 algorithm, NASA GISS

Software: three.js (MIT), numpy, scipy, rasterio / GDAL, Pillow, pywebview, pythonnet and PyInstaller. Fonts in this guide: Barlow and IBM Plex Mono (SIL Open Font License). MarsNav's own code is released under the MIT licence (LICENSE).

About this guide

Written for MarsNav 2.1 for the NASA Space Apps Challenge 2026 (Interplanetary Survival Guide: Martian Map). Every screenshot was taken from the running app on real data: the 2.0 figures in the desktop window, the 2.1 figures (Chapters 4-6, 8, 10 and 13) in a browser on the same app, without the window frame. The validation numbers in Chapter 16 were re-run for 2.0 on 26 October 2026; the tests, the Windows build and the browser tour for 2.1 on 29 October 2026.